MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, DIGITAL, KEY CODE KYF

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital random access memory module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Memory, random access
Family	Digital
Key code	KYF
Size	1A
Weight	35 grams maximum
Failure rate	11.665 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10 percent
Maximum power dissipation	
All chips enabled	3.82 W (outputs not loaded)
One chip enabled	1.52 W (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, A0-A11	
Low level (V_{IL} = 0.0 V)	10 µA maximum
(V_{IL} = 0.4 V)	10 µA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	10 µA maximum
Input current, $\overline{WE}0$-$\overline{WE}3$, DI0-DI3	
Low level (V_{IL} = 0.0 V)	10 µA maximum
(V_{IL} = 0.4 V)	10 µA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	10 µA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Input current, $\overline{CS0}$-$\overline{CS3}$	
Low level (V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	50 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input clamp diode voltage, $\overline{CS0}$-$\overline{CS3}$ (V_{CC} = 4.5 V, I_{clamp} = -12 mA)	-1.5 V maximum
Output voltage	
Low level (I_{OL} = 16 mA)	0.4 V maximum
High level (I_{OH} = -2 mA)	2.4 V minimum
Output disabled current	
Low level (V_{OL} = 0.4 V)	±40 µA maximum
High level (V_{OH} = 2.4 V)	±40 µA maximum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	16 mA
High level (V_{OH} = 2.4 V)	-2 mA
Input capacitance	
Data-in, DI0-DI3	15 pF maximum
Address, A0-A11	45 pF maximum
Circuit-select (enable), $\overline{CS0}$-$\overline{CS3}$	25 pF maximum
Write-enable, $\overline{WE0}$-$\overline{WE3}$	15 pF maximum
Output capacitance, DO0-DO3	25 pF maximum
Dynamic characteristics (see figure 1)	
Setup time	
t_{AS} - address to write-enable	150 ns minimum
t_{SS} - circuit-select (low-to-high) to write-enable (high-to-low) to prevent writing	50 ns minimum
t_{SS} - circuit-select (high-to-low) to write-enable (low-to-high) to assure writing	300 ns minimum
t_{DS} - data-in to write-enable (low-to-high)	300 ns minimum
Hold time	
t_{AH} - write-enable (low-to-high) to address	100 ns minimum
t_{SH} - write-enable (low-to-high) to circuit select	50 ns minimum
t_{DH} - write-enable (low-to-high) to data	100 ns minimum
Write pulse width (t_{WP})	300 ns minimum
Address access time (t_{AA})	
C_L = 50 pF	525 ns maximum
C_L = 150 pF	530 ns maximum
Select access time (t_{SA})	
C_L = 50 pF	375 ns maximum
C_L = 150 pF	380 ns maximum
Select recovery time	
t_{SRL} - low state to high impedance state ($C_L \leq 10$ pF)	40 ns maximum
t_{SRH} - high state to high impedance state ($C_L \leq 10$ pF)	40 ns maximum
Write recovery time (t_{WR})	

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C_L = 50 pF	100 ns maximum
C_L = 150 pF	110 ns maximum

1.3 Module function. This module can be used in any of three read/write random access memory configurations:

a. 4096 words by 4 bits.
b. 8192 words by 2 bits.
c. 16384 words by 1 bit.

For the 4096-word by 4-bit memory configuration, all four write-enable inputs ($\overline{WE}$0-$\overline{WE}$3) and select inputs ($\overline{CS}$0-$\overline{CS}$3) are tied together at the module connector, whereas the four data inputs (DI0-DI3), and the four data outputs (DO0-DO3) are all independent. Functionally, the module is as shown on figure 2.

If a 8192-word by 2-bit memory configuration is required, two dual 4096-word by 2-bit groups are formed by pairing circuit-selects, write-enables, data inputs, and data outputs using the procedure outlined below:

Any two of the four circuit-select ($\overline{CS}$) signals can be tied together; the remaining two must form the other pair. Then the write-enable inputs ($\overline{WE}$), data inputs (DI) and data outputs (DO) of one of the two devices whose $\overline{CS}$'s are tied together are paired with the $\overline{WE}$, DI, and DO of either of the two devices that form the other pair. Finally, the $\overline{WE}$, DI, and DO of the remaining two devices are paired. Examples are given below:

Group A pairs	Group B pairs
$\overline{CS}$0 and $\overline{CS}$2	$\overline{CS}$1 and $\overline{CS}$3
$\overline{WE}$0 and $\overline{WE}$3	$\overline{WE}$1 and $\overline{WE}$2
DI0 and DI3	DI1 and DI2
DO0 and DO3	DO1 and DO2

Functionally, the module is as shown on figure 3.

A 16384-word by 1-bit memory is achieved by tying together the four write-enable inputs ($\overline{WE}$), the four data inputs (DI), and four data outputs (DO), respectively. The four circuit-select inputs remain independent. Functionally, the module is as shown on figure 4.

1.3.1 Input to output relationship. The input to output relationship for read and write operations are described below and in table I and figure 1.

1.3.1.1 Read operation. The memory is addressed by the address inputs (A0-A11) which are connected in parallel to all four RAM circuits and which decode one of the 4096 storage cells in each of the four devices. When the appropriate circuit-selects ($\overline{CS}$0-$\overline{CS}$3) go to the low level, data stored in the selected locations of the enabled circuits is read out on the data-out pins. Each of the four data-outs is buffered by a tri-state gate on the module. The data read out has the same polarity as the data written in during the previous write cycle.

1.3.1.2 Write operation. The memory is addressed by the A0-A11 inputs which are connected in parallel to all four RAM devices and which decode to one of the 4096 storage cells in each of the four devices. The circuits which are to be written into are enabled by the appropriate low-going circuit-select and write-enable inputs. The data present on the data input pins of the enabled circuits is written into the selected location.

1.3.2 Paralleling of inputs and outputs. In addition to the paralleling of signals on a single module already discussed in 1.3, bussing of like signals between modules is allowed to achieve expanded memory capability. For example, the address terms, data-in, write-enable, and data-out signals of two modules may be connected together respectively to form a 32768-word by 1-bit configuration. A number of memory outputs may be connected to a common bus. All of the common outputs except one are placed in the high impedance state and one output is selected. Two or more common outputs may not be selected simultaneously.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION

MILITARY

3202946G1 - SEM Memory Module Test Adapter.

(All Ordnance Data publications shall be ordered from the Commanding Officer, Technical Document Department, Naval Ordnance Station, Louisville, KY 40214.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 11.665 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KYF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

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3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/286-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 5.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for test condition B (V_{CC} stress test) apply only for a power supply voltage of 7.0 Vdc +0, -1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	2
Megohm bridge	Item R06	1
DC ammeter	Item X01	1
DC voltmeter	Item V03	1
Resistance bridge	Item R02	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
LSI tester with SEM memory module adapter dwg No. 3202946G1	Item U03	1
Capacitance meter	Item C05	1
Capacitance bridge	Item C01	1
Resistors		
274Ω ±1%, 1/4 W		5
51Ω ±5%, 1/4 W		1
2.2 kΩ ±5%, 1/8 W		3
100Ω ±1%, 1/8 W		1
887Ω ±1%, 1/4 W		1
1.2 kΩ ±5%, 1/8 W		2
51Ω ±5%, 1/2 W		2
1.2 kΩ ±1%, 1/8 W		4
51Ω ±5%, 1 W		3
150Ω ±5%, 1/4 W		1
1.0Ω ±1%, 1/2 W		1
Values as required for figure 18, note 2, 1/8 W		4

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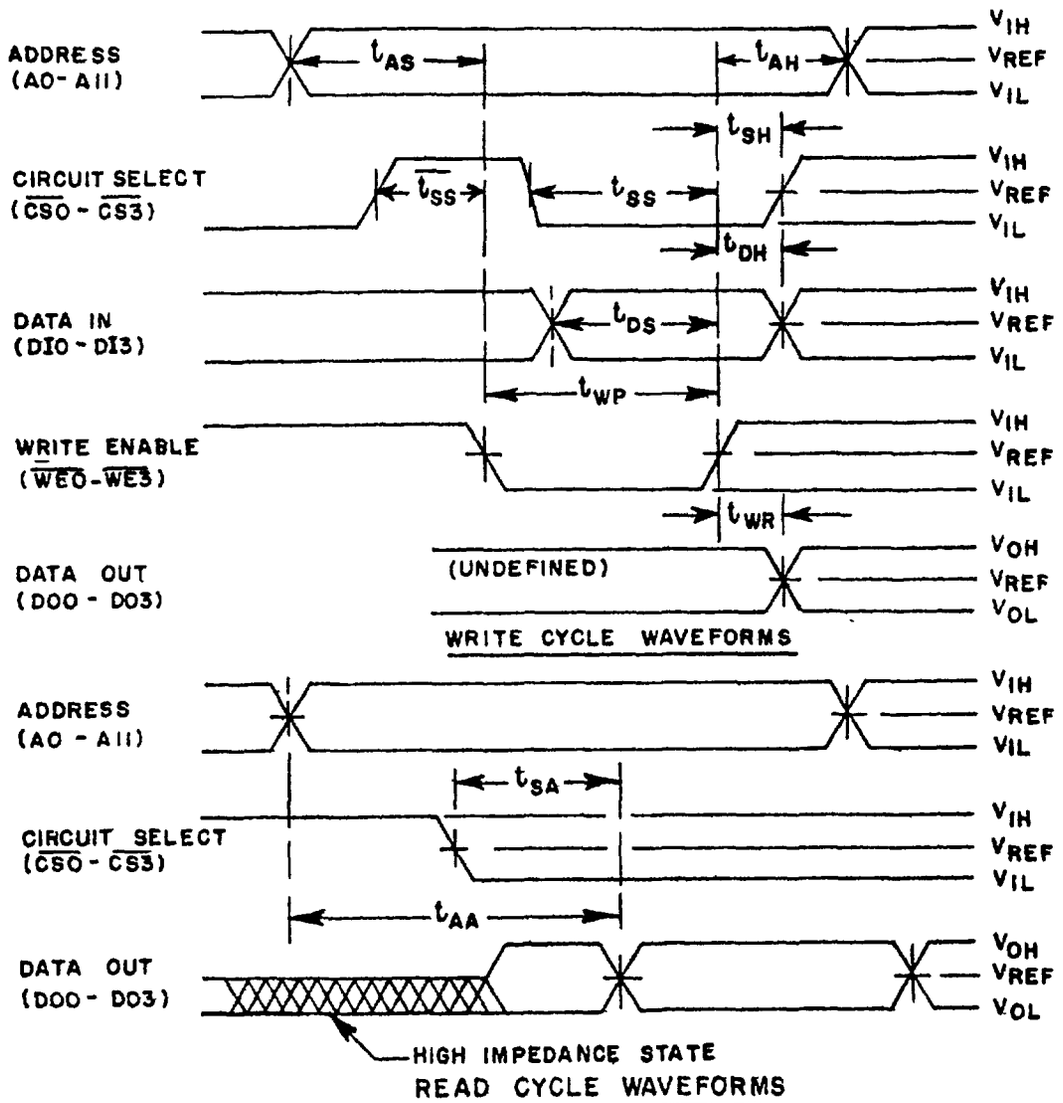
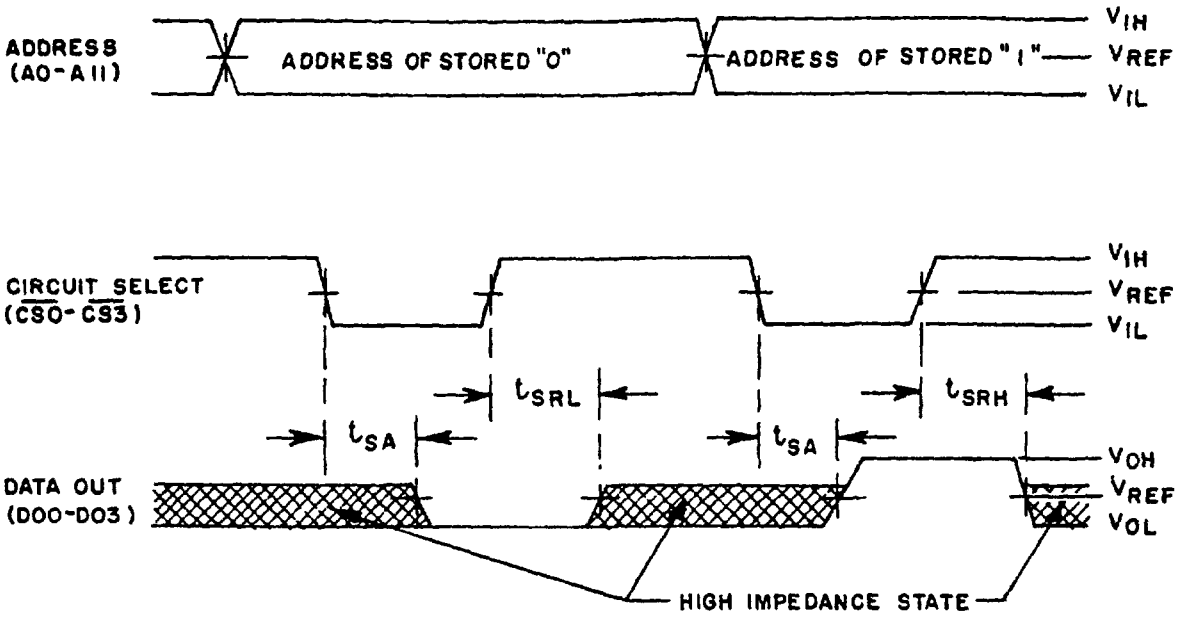


FIGURE 1. KYF dynamic characteristics waveforms.

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ENABLE / DISABLE CYCLE WAVEFORMS

FIGURE 1. KYF dynamic characteristics waveforms - Continued.

TABLE I. Input to output relationship for each RAM.

Inputs			Output	Mode
$\overline{CS}$	$\overline{WE}$	DI	DO	
H	X	X	High Z	Disabled
L	L	H	Undefined	Write high
L	L	L	Undefined	Write low
L	H	X	Data in	Read data from addressed location

NOTES:
1. H = High level
2. L = Low level
3. X = Don't care
4. High Z = High-impedance state

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature 4/				Operating temperature 0° to 60°C 4/				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Frame isolation	III	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
A2-A10	Unused pin isolation	III	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
B	V_{CC} stress	IV	Q	← See table IV →								
C1-C4	Input breakdown current, $\overline{CS0}$ - $\overline{CS3}$	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C5-C16	Input leakage current, A0-A11	V	X	0	10	0	10	0	10	0	10	µA
C17-C24	Input leakage current, $\overline{WE0}$-$\overline{WE3}$, DI0-DI3	V	X	0	10	0	10	0	10	0	10	µA
D1-D4	Output leakage current, low level outputs	VI	X	-40	40	-40	40	-40	40	-40	40	µA
E1-E4	Output leakage current, high level outputs	VI	X	-40	40	-40	40	-40	40	-40	40	µA
F1-F4	Output short circuit current	VII	X	-30	-70	-30	-70	-30	-70	-30	-70	mA
G1-G4	Input clamp diode, $\overline{CS0}$-$\overline{CS3}$	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H4	Static output voltage, low level	IX	X	0.01	0.4	0.01	0.4	0.01	0.4	0.01	0.4	V
J1-J4	Static output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K4	Input current, high level, $\overline{CS0}$-$\overline{CS3}$	X	X	0	50	0	50	0	50	0	50	µA
L1-L4	Input current, low level, $\overline{CS0}$-$\overline{CS3}$	XI	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA

See footnotes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature 4/				Operating temperature 0° to 60°C 4/				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L5-L16	Input current, low level, A0-A11	XI	X	-10	1	-10	1	-10	1	-10	1	µA
L17-L24	Input current, low level, $\overline{WE0}$-$\overline{WE3}$, DI0-DI3	XI	X	-10	1	-10	1	-10	1	-10	1	µA
M1-M4	Address input threshold check, low level	XII	X	0	0.40	0	0.40	0	0.40	0	0.40	V
M5-M8	Address input threshold check, high level	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
N1-N2	Ground and V_{CC} pin continuity	XIII	X	0	1.0	0	1.0	0	1.0	0	1.0	Ω
N3-N4	Ground and V_{CC} pin continuity	XIII	Q	0	0.1	0	0.1	0	0.1	0	0.1	Ω
P1	Power supply current	XIV	X	50	694	50	694	50	694	50	694	mA
P2-P6	Power supply current	XIV	X	50	275	50	275	50	275	50	275	mA
Q1-Q4	Transition time (t_{TLH}), C_L = 50 pF	XV	S	5	25	5	25	5	25	5	25	ns
Q5-Q8	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	5	40	5	40	5	40	5	40	ns
Q9-Q12	Transition time (t_{THL}), C_L = 50 pF	XV	S	3	15	3	15	3	15	3	15	ns
Q13-Q16	Transition time (t_{THL}), C_L = 150 pF	XV	Q	3	25	3	25	3	25	3	25	ns

See footnotes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature 4/				Operating temperature 0° to 60°C 4/				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R1-R4	Select recovery time (t_{SRL})	XVI	S	5	40	5	40	5	40	5	40	ns
R5-R8	Select recovery time (t_{SRH})	XVI	S	5	40	5	40	5	40	5	40	ns
S1-S6	Dynamic test, C_L = 50 pF	XVII	X	← No fail →								
S7-S12	Dynamic tests, C_L = 150 pF	XVII	Q	← No fail →								
T1-T8	Input crosstalk, low level	XVIII	Q	← See note 4 and table XVIII →								
T9-T16	Input crosstalk, high level	XVIII	Q	← See note 4 and table XVIII →								
U	Power supply transient amplitude	XIX	Q	0	700	0	700	0	700	0	700	mV
V	Power supply filter capacitance	XX	S	3.9	7.3	3.9	7.3	3.9	7.3	3.9	7.3	µF
W1-W4	Input capacitance CS0-CS3	XXI	Q	2.0	25.0	2.0	25.0	2.0	25.0	2.0	25.0	pF
W5-W16	Input capacitance, A0-A11	XXI	Q	2.0	45.0	2.0	45.0	2.0	45.0	2.0	45.0	pF
W17-W20	Input capacitance, $\overline{WE0}$-$\overline{WE3}$	XXI	Q	2.0	15.0	2.0	15.0	2.0	15.0	2.0	15.0	pF
W21-W24	Input capacitance, DI0-DI3	XXI	Q	2.0	15.0	2.0	15.0	2.0	15.0	2.0	15.0	pF
X1-X4	Output capacitance, DO0-DO3	XXI	Q	2.0	25.0	2.0	25.0	2.0	25.0	2.0	25.0	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Initial and end-of-life limits for acceptance and operating requirements are the same.

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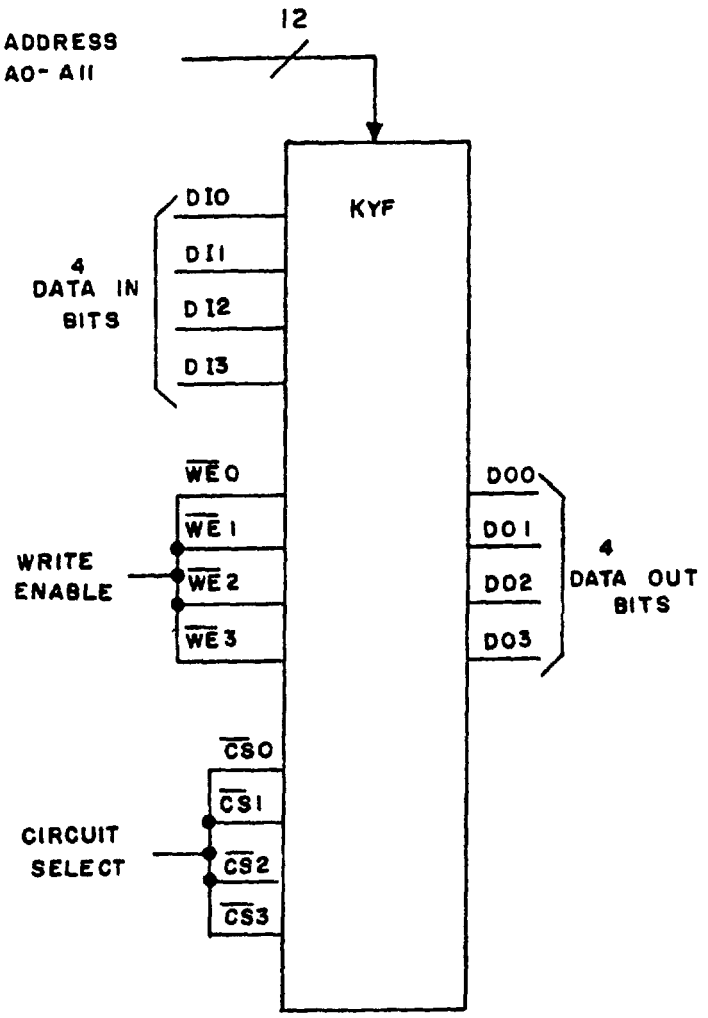


FIGURE 2. 4096-word by 4-bit configuration.

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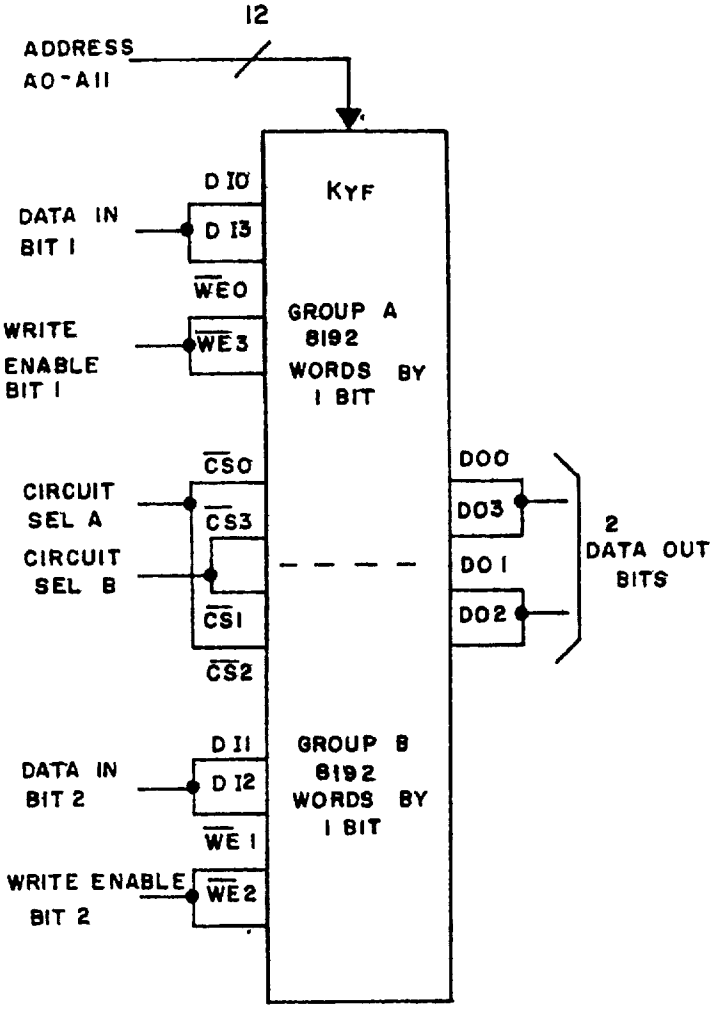


FIGURE 3. 8192-word by 2-bit configuration.

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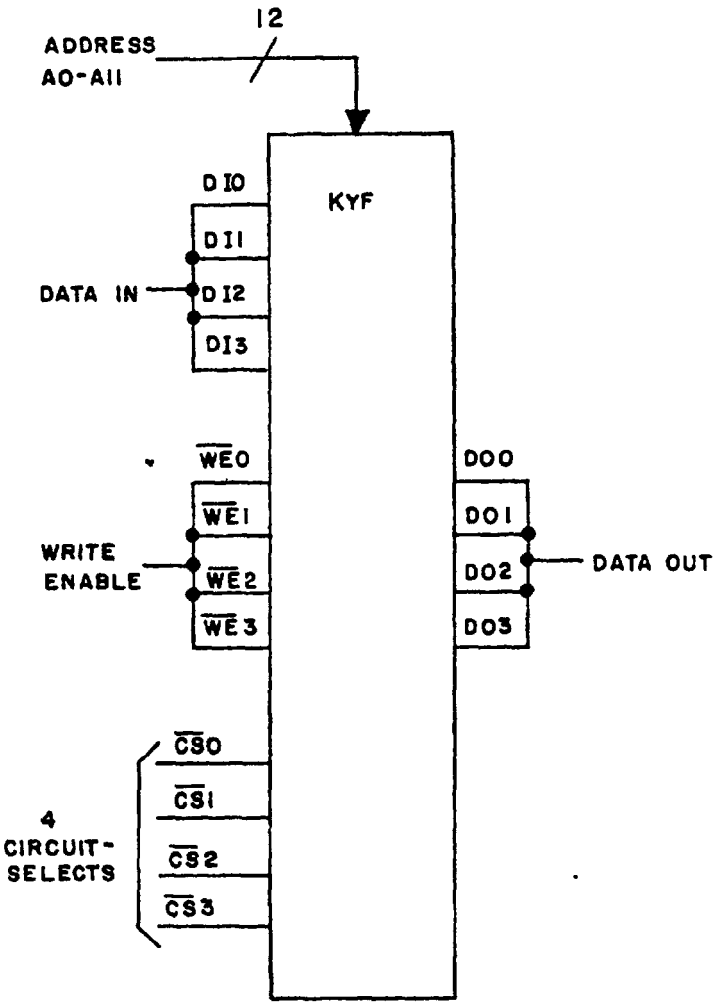
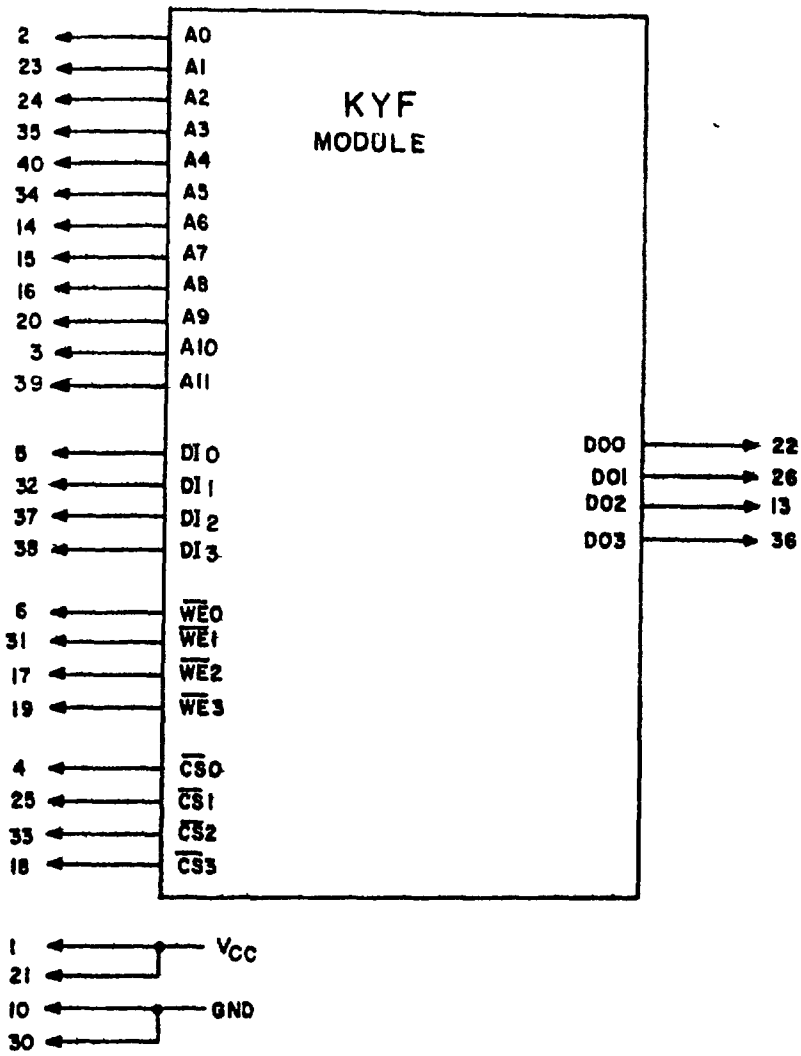


FIGURE 4. 16384-word by 1-bit configuration.

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NOTES:
1. A0-A11 = Address inputs.
2. DI0-DI3 = Data inputs.
3. DO0-DO3 = Data outputs.
4. $\overline{WE0}$-$\overline{WE3}$ = Write-enable inputs.
5. $\overline{CS0}$-$\overline{CS3}$ = Circuit-select inputs.
6. Unused pins: 7, 8, 9, 11, 12, 27, 28, 29.
7. V_{CC} = 5.0 V ±10%.
8. Ground = 0 V reference.

FIGURE 5. KYF function diagram, 16384-bit random access memory.

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Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
Capacitors		
Value as required for figure 18, note 4		4
Value as required for figure 18, note 5		4
Value as required for figure 18, note 6		4
Value as required for figure 32, Note 6		As required
Diodes		
1N3600, 1N4150, or equivalent		4
Integrated circuits		
9316		4
5400		1
Switches		
SPDT		1
SPST		1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions:

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 6.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 9.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 10. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 6. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C,

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electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

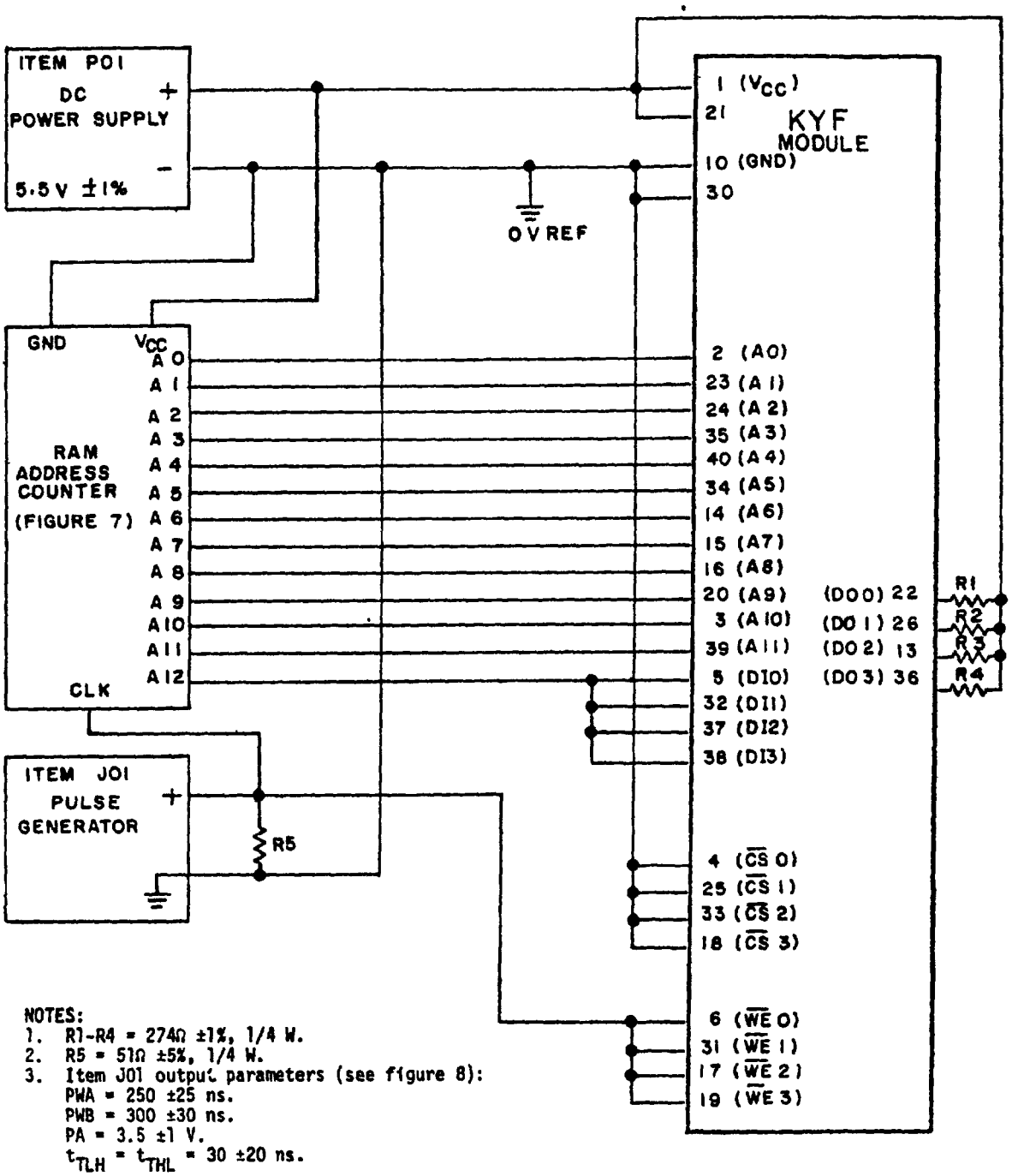
6.3 Handling and storage precautions. To avoid possible damage to devices during handling, testing, or actual operation, the following procedures should be followed: Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid buildup of static charges; soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N004)

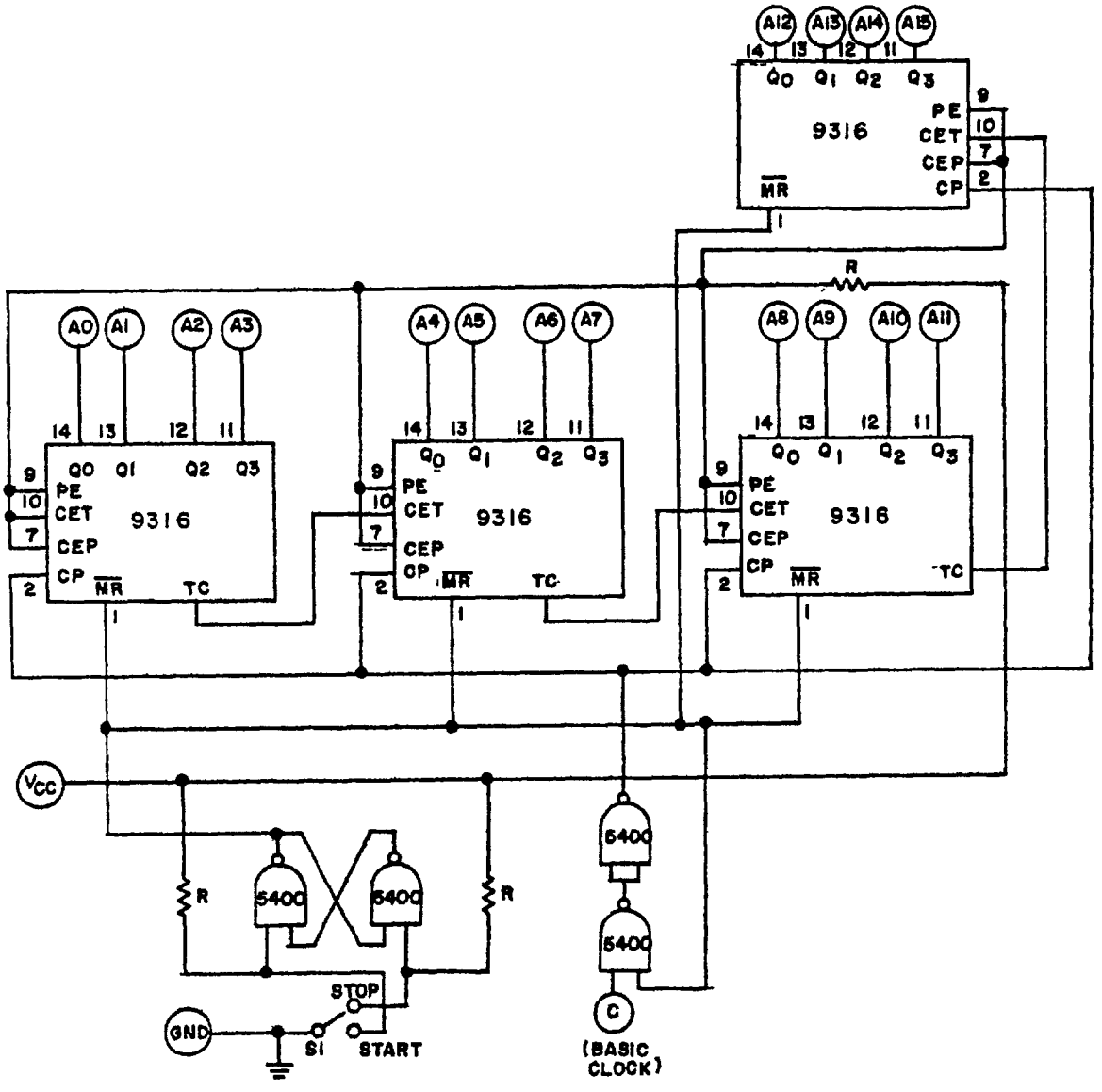
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NOTES:
1. R1-R4 = 274Ω ±1%, 1/4 W.
2. R5 = 51Ω ±5%, 1/4 W.
3. Item J01 output parameters (see figure 8):
   PWA = 250 ±25 ns.
   PWB = 300 ±30 ns.
   PA = 3.5 ±1 V.
   $t_{TLH} = t_{THL}$ = 30 ±20 ns.

FIGURE 6. Operating temperature and life test circuit.

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NOTES:
1. 3 each R = 2.2 kΩ ±5%, 1/8 W.
2. 4 each 9316 high-speed synchronous 4-bit binary counters.
3. 1 each 5400 quad 2-input NAND gate.
4. Switch S1 = SPDT.

FIGURE 7. RAM address counter.

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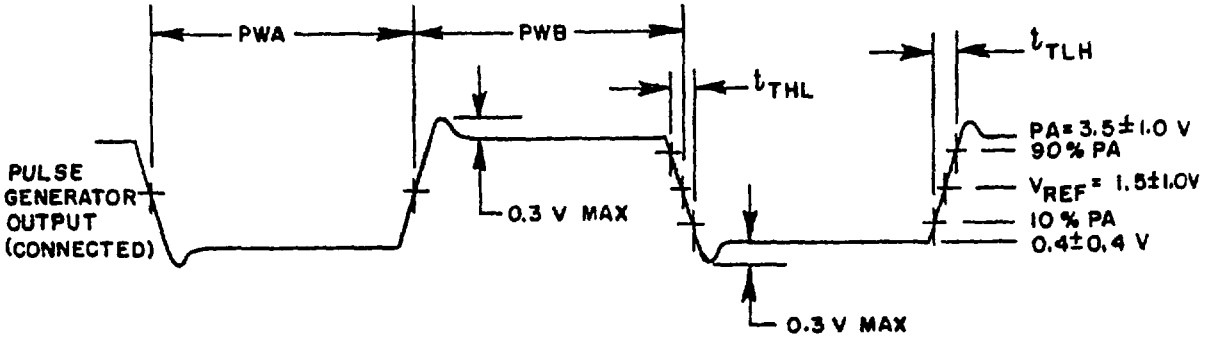
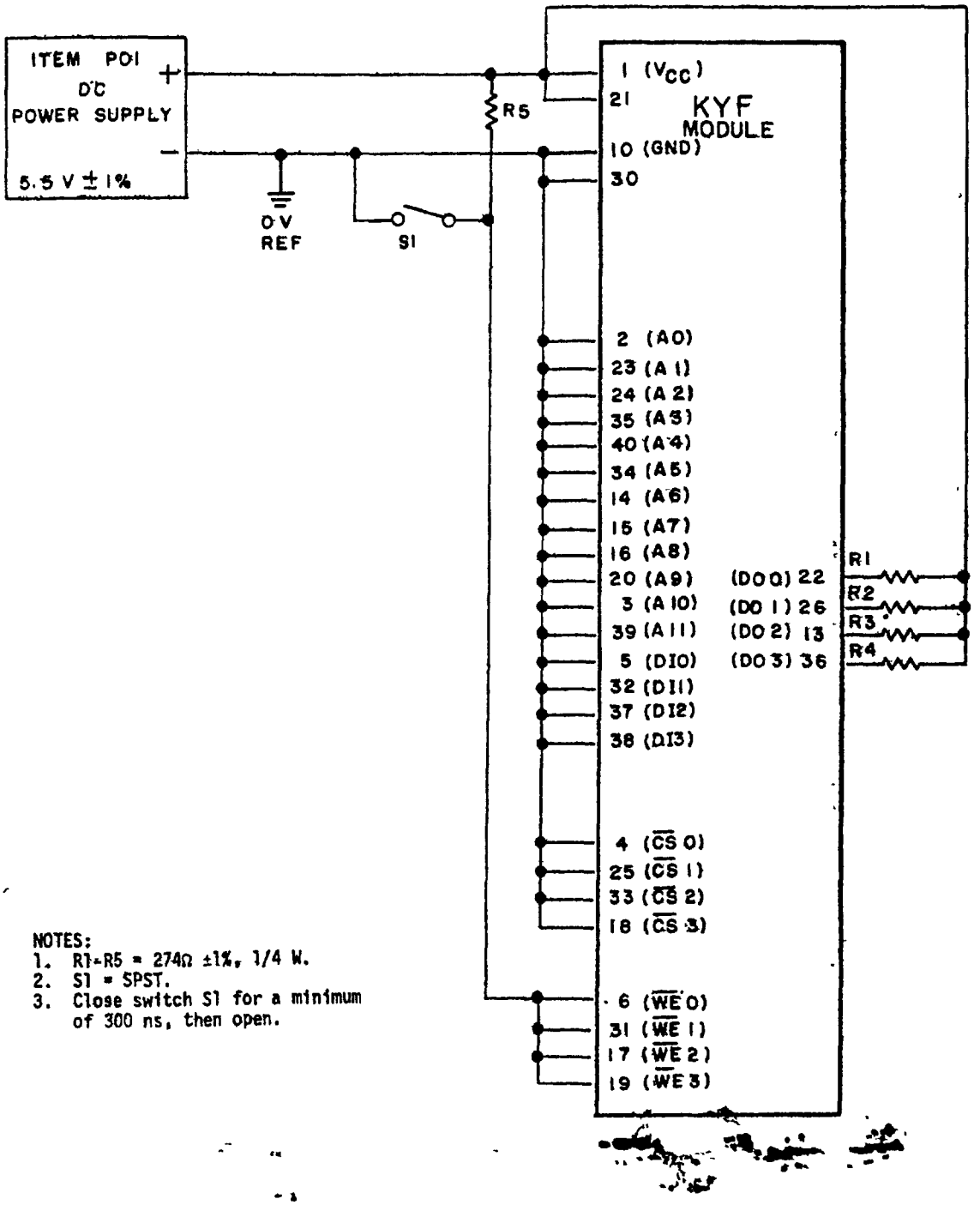


FIGURE 8. Operating temperature and life test waveform.

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NOTES:
1. R1-R5 = 274Ω ±1%, 1/4 W.
2. S1 = SPST.
3. Close switch S1 for a minimum of 300 ns, then open.

FIGURE 9. Component temperature test circuit.

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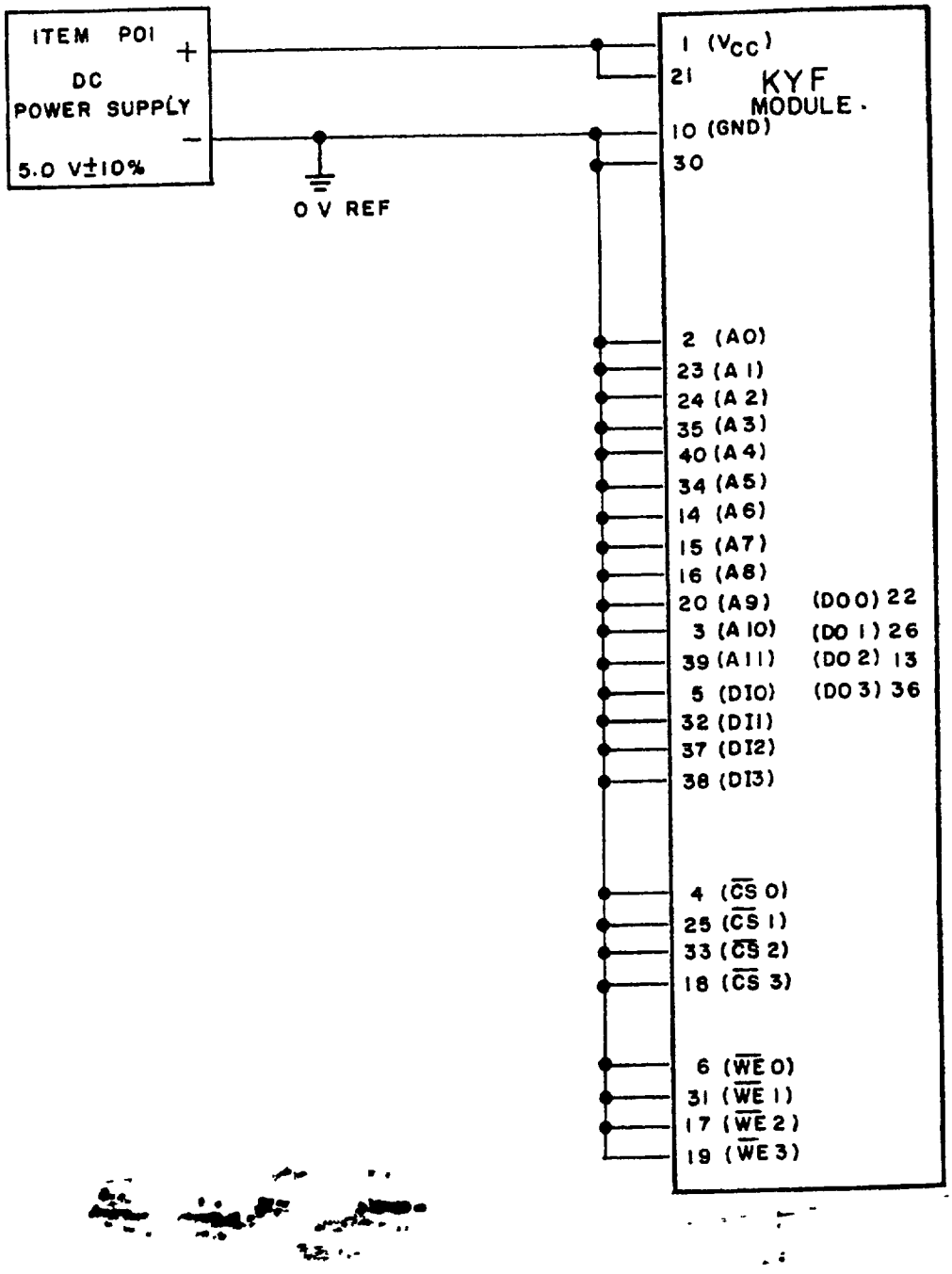


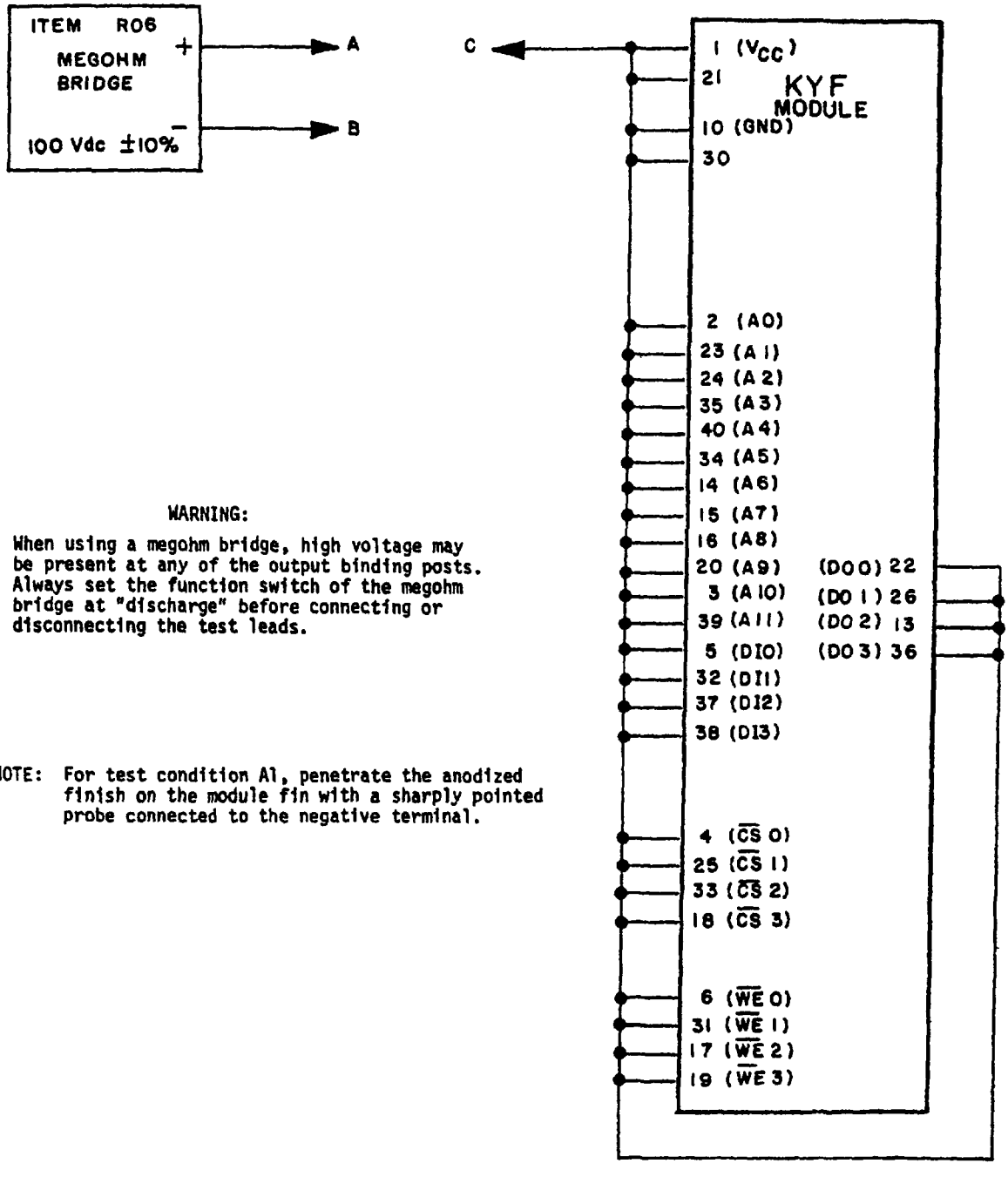
FIGURE 10. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Frame isolation	C, 7,8,9,11,12, 27,28,29	Module fin	10	∞	MΩ	11
A2	Unused pin isolation	C	7,8,9,11,12, 27,28,29				
A3		7	8,9,11,12, 27,28,29				
A4		8	7,9,11,12, 27,28,29				
A5		9	7,8,11,12, 27,28,29				
A6		11	7,8,9,12, 27,28,29				
A7		12	7,8,9,11, 27,28,29				
A8		27	7,8,9,11,12, 28,29				
A9		28	7,8,9,11,12, 27,29				
A10	Unused pin isolation	29	7,8,9,11,12, 27,28	10	∞	MΩ	11

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module fin with a sharply pointed probe connected to the negative terminal.

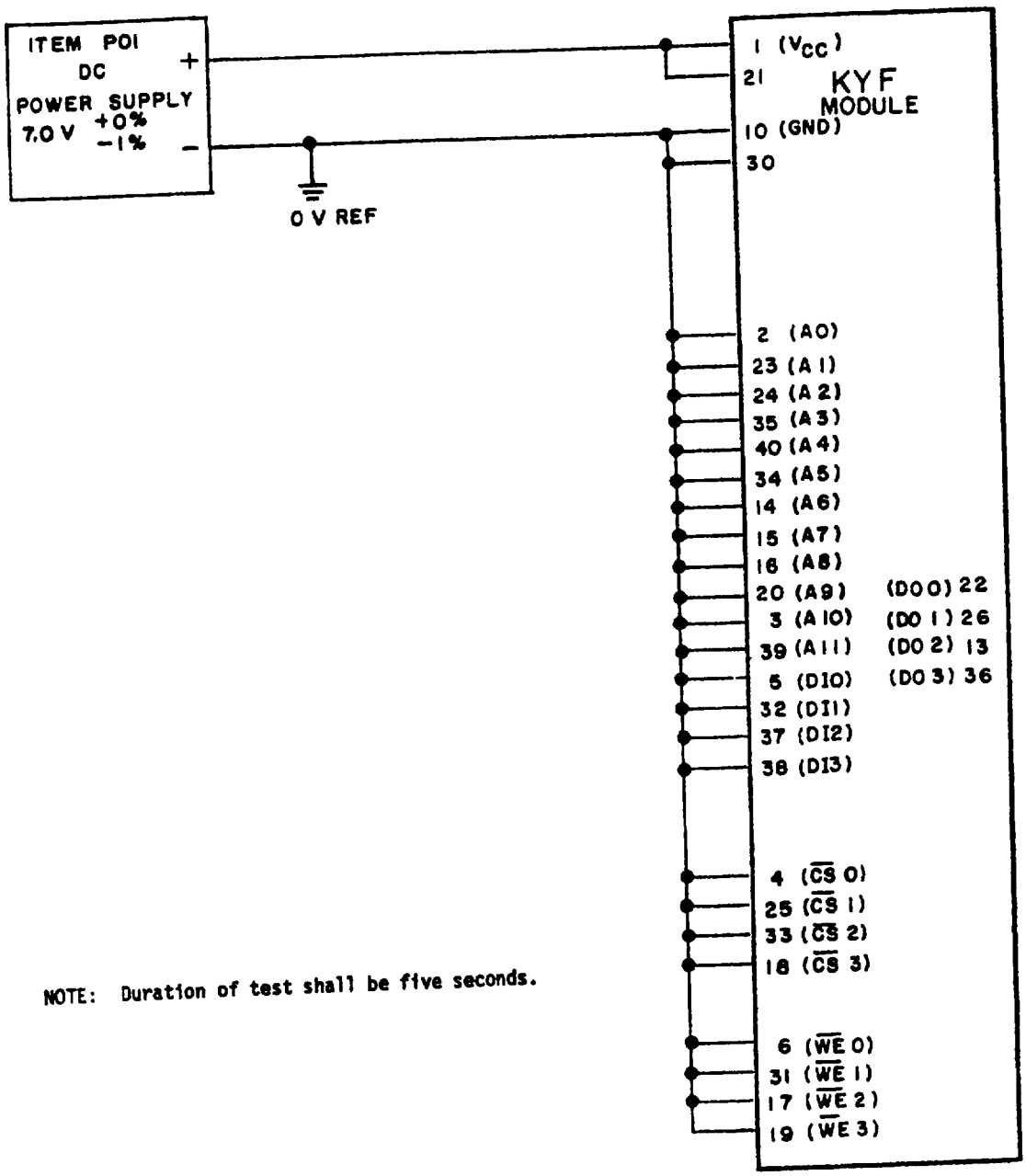
FIGURE 11. Isolation test circuit for table III.

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TABLE IV. V_{CC} stress test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	Must pass remainder of tests			12

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NOTE: Duration of test shall be five seconds.

FIGURE 12. V_{CC} stress test circuit for table IV.

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TABLE V. Input breakdown current and input leakage current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	$\overline{CS0}$ Input breakdown current	2,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,35, 37,38,40,39	4	0.0	1.0	mA	13
C2	$\overline{CS1}$	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,31, 32,33,34,35, 37,38,40,39	25				
C3	$\overline{CS2}$	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,34,35,37,38 40,39	33				
C4	$\overline{CS3}$ Input breakdown current	2, 4, 5,6,14, 15,16,17,3,19, 20,23,24,25,31 32,33,34,35, 37,38,40,39	18	0.0	1.0	mA	
C5	A0 Input leakage current	4,5,6,14,15,16 17,3,18,19,20, 23,24,25,31,32 33,34,35,37,38 40,39	2	0	10	µA	
C6	A1 Input leakage current	2,4,5,6,14,15, 16,17,3,18,19, 20,24,25,31, 32,33,34,35, 37,38,40,39	23	0	10	µA	13

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TABLE V. Input breakdown current and input leakage current test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C7	A2 Input leakage current	2,4,5,6,14,15, 16,17,3,18,19, 20,23,25,31,32, 33,34,35,37,38, 40,39	24	0	10	µA	13
C8	A3	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,37,38, 40,39	35				
C9	A4	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,35,37, 38,39	40				
C10	A5	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,35,37, 38,40,39	34				
C11	A6	2,4,5,6,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37, 38,40,39	14				
C12	A7 Input leakage current	2,4,5,6,14,16, 17,3,18,19,20, 23,24,25,31, 32,33,34,35, 37,38,40,39	15	0	10	µA	13

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TABLE V. Input breakdown current and input leakage current test -Continued.

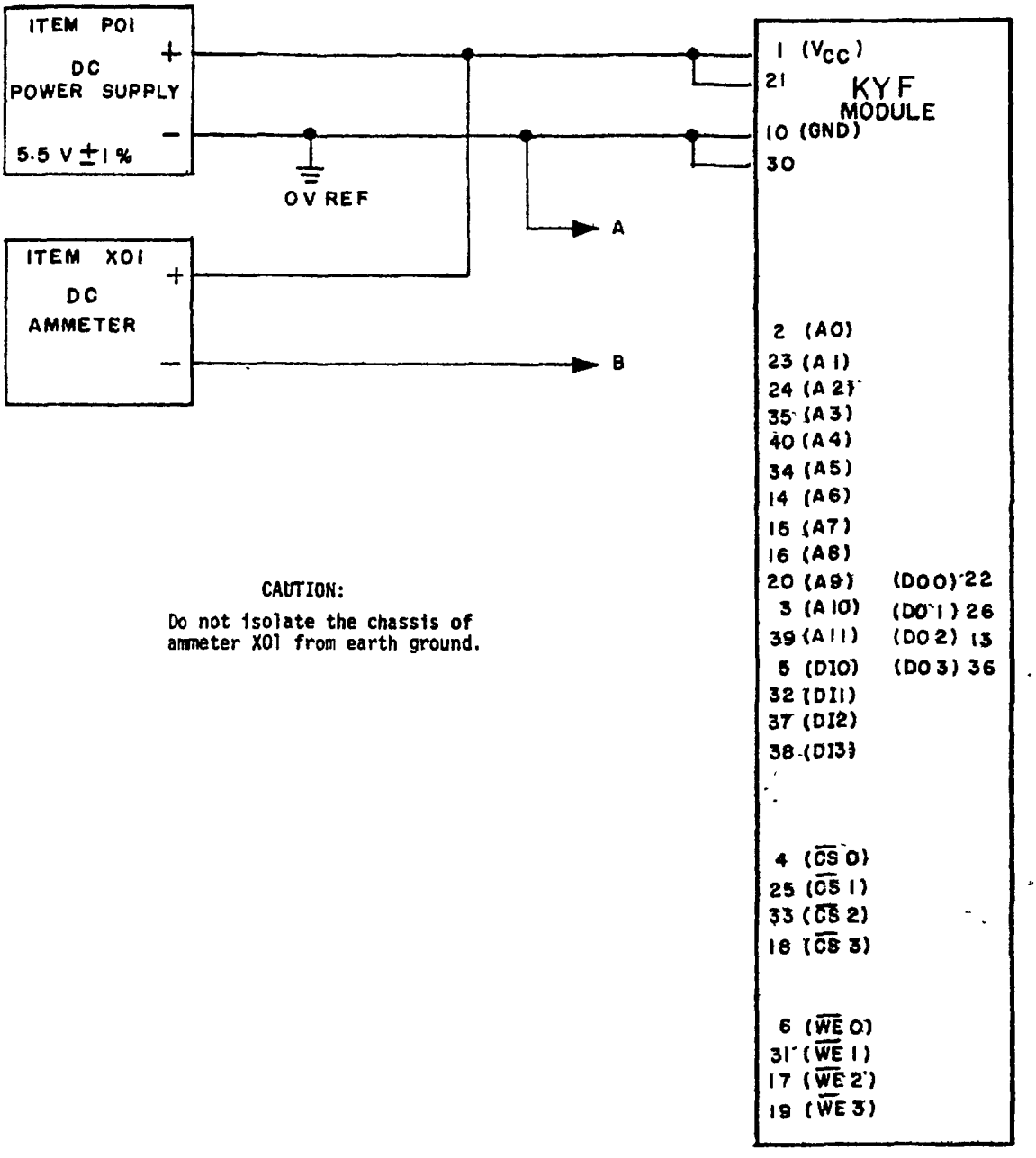
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C13	A8 Input leakage current	2,4,5,6,14,15, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37, 38,40,39	16	0	10	µA	13
C14	A9	2,4,5,6,14,15, 16,17,3,18,19, 23,24,25,31,32, 33,34,35,37,38, 40,39	20				
C15	A10	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	3				
C16	A11	2,3,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40	39		10		
C17	$\overline{WE0}$	2,4,5,14,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	6		10		
C18	$\overline{WE1}$ Input leakage current	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,32, 33,34,35,37, 38,40,39	31	0	10	µA	13

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TABLE V. Input breakdown current and input leakage test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	$\overline{WE2}$ Input leakage current	2,4,5,6,14,15, 16,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	17	0	10	µA	13
C20	$\overline{WE3}$	2,4,5,6,14,15, 16,17,3,18, 20, 23,24,25,31,32, 33,34,35,37,38, 40,39	19				
C21	DI0	2,4,6,14,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	5				
C22	DI1	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 33,34,35,37,38, 40,39	32				
C23	DI2	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,35,38, 40,39	37				
C24	DI3 Input leakage current	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,35,37, 40,39	38	0	10	µA	13

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

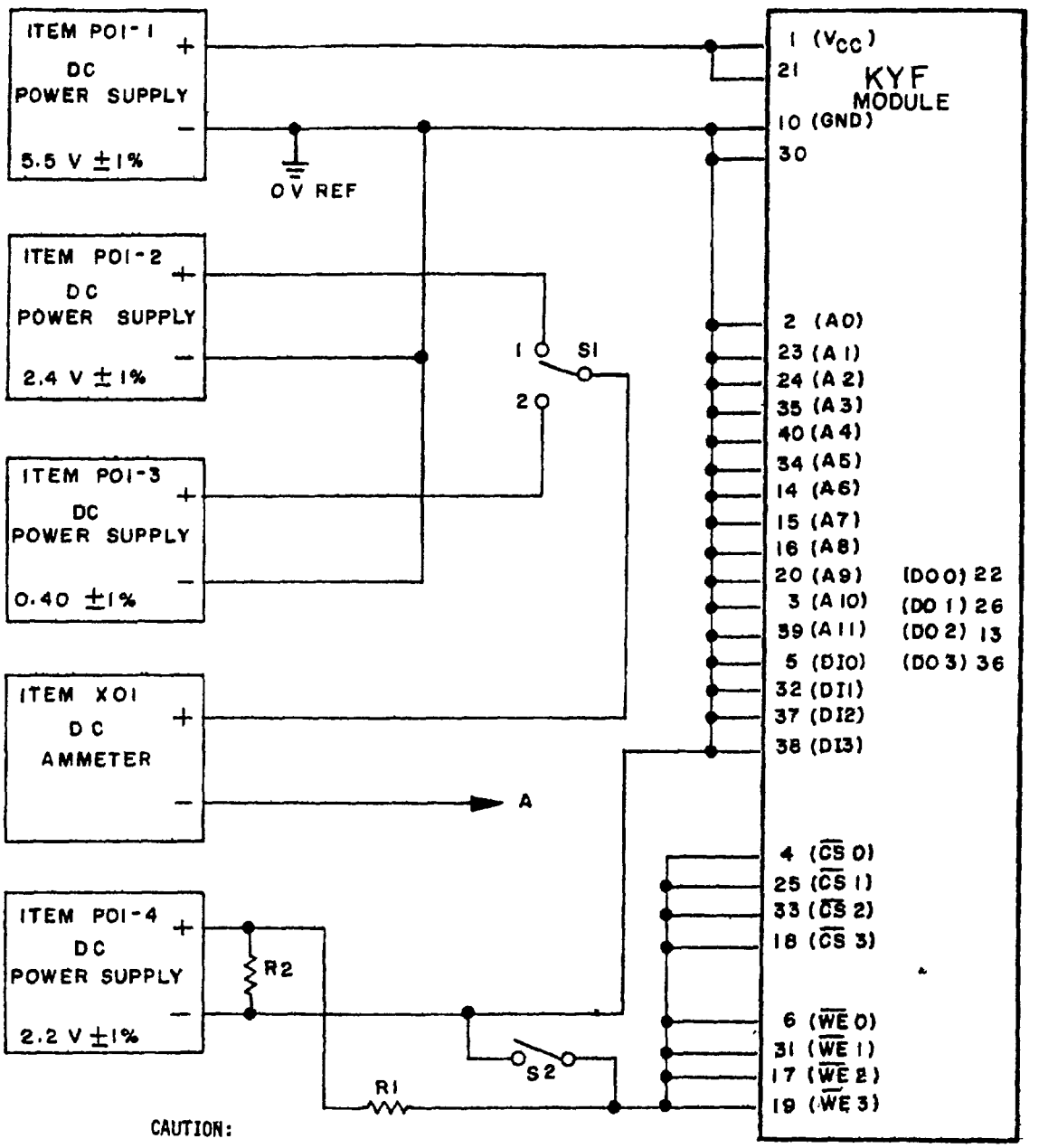
FIGURE 13. Input breakdown current and input leakage current test circuit for table V.

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TABLE VI. Output leakage current (low-level and high-level outputs) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	DO0 Output leakage current, low level outputs	22	-40	40	µA	14
D2	DO1	26				
D3	DO2	13				
D4	DO3 Output leakage current, low level outputs	36				
E1	DO0 Output leakage current, high level outputs	22				
E2	DO1	26				
E3	DO2	13				
E4	DO3 Output leakage current, high level outputs	36	-40	40	µA	14

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Switch S1 in position 2 for test conditions D1-D4.
2. Switch S1 in position 1 for test conditions E1-E4.
3. R1 = 100Ω ±1%, 1/8 W.
4. R2 = 51Ω ±5%, 1/4 W.
5. Before performing a test, close switch S2 for a minimum of 500 ns, then open and make measurements.
6. S1 = SPDT.
7. S2 = SPST.

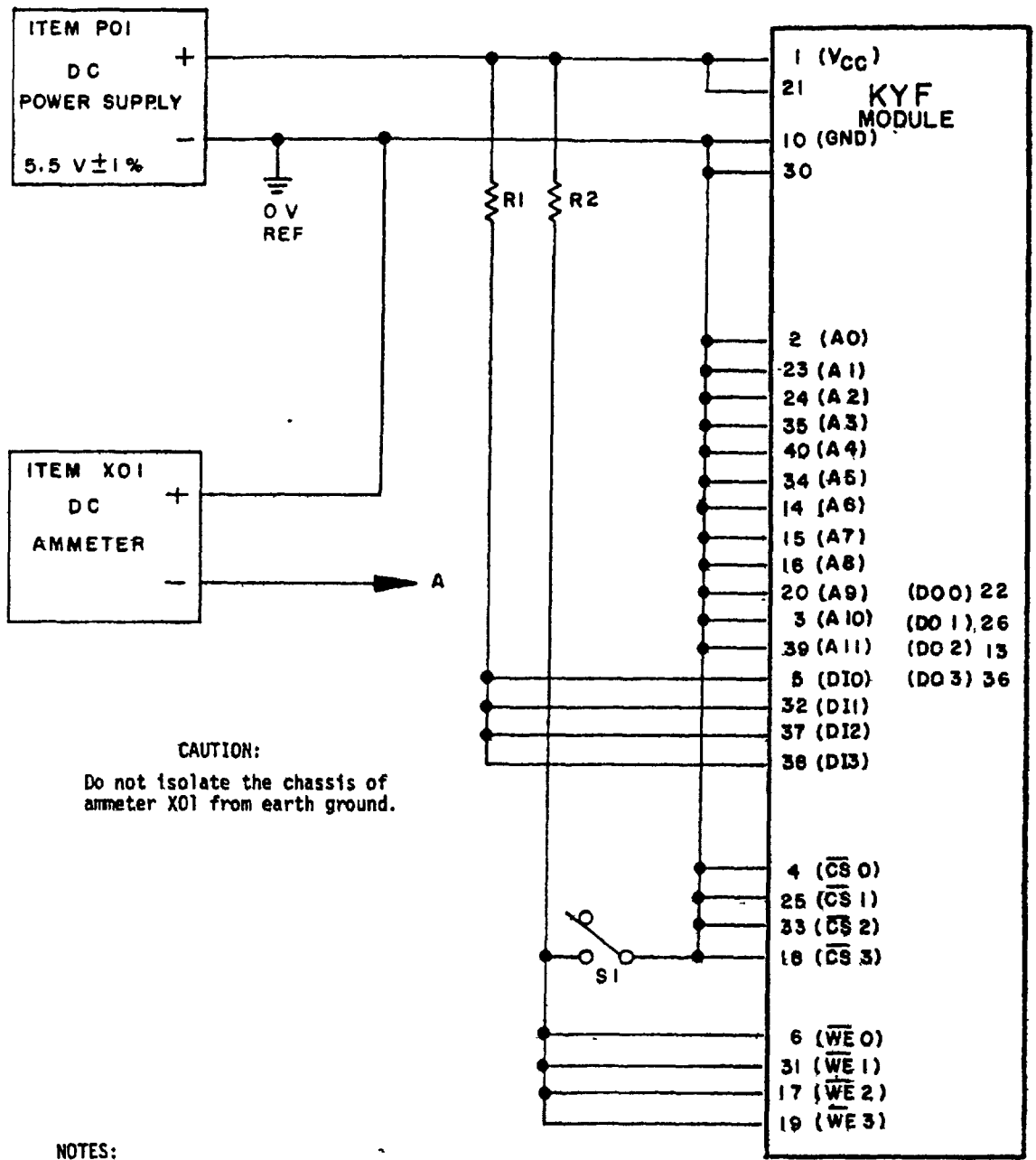
FIGURE 14. Output leakage current (low-level and high-level outputs) test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F1	000 Output short circuit current	22	-30	-70	mA	15
F2	001	26	↓	↓	↓	↓
F3	002	13	↓	↓	↓	↓
F4	003 Output short circuit current	36	-30	-70	mA	15

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NOTES:
1. When performing output short circuit tests, close switch 1 for a minimum of 500 ns, then open.
2. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed ten seconds.
3. R1, R2 = 1.2 kΩ ±5%, 1/8 W.
4. S1 = SPST.

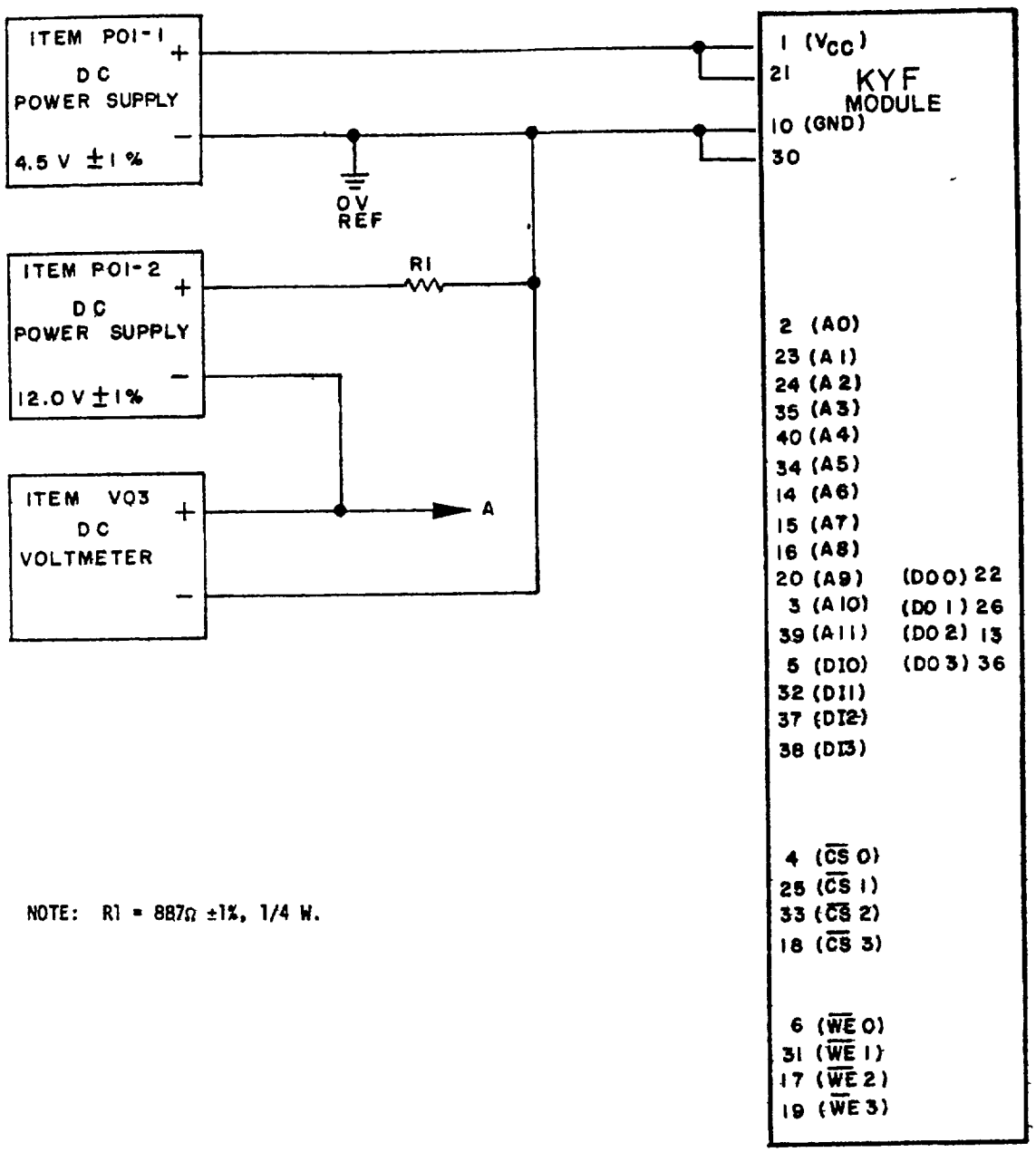
FIGURE 15. Output short circuit current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	$\overline{CS0}$ Input clamp diode voltage	4	-0.4	-1.5	V	16
G2	$\overline{CS1}$	25				
G3	$\overline{CS2}$	33				
G4	$\overline{CS3}$ Input clamp diode voltage	18	-0.4	-1.5	V	16

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NOTE: R1 = 887Ω ±1%, 1/4 W.

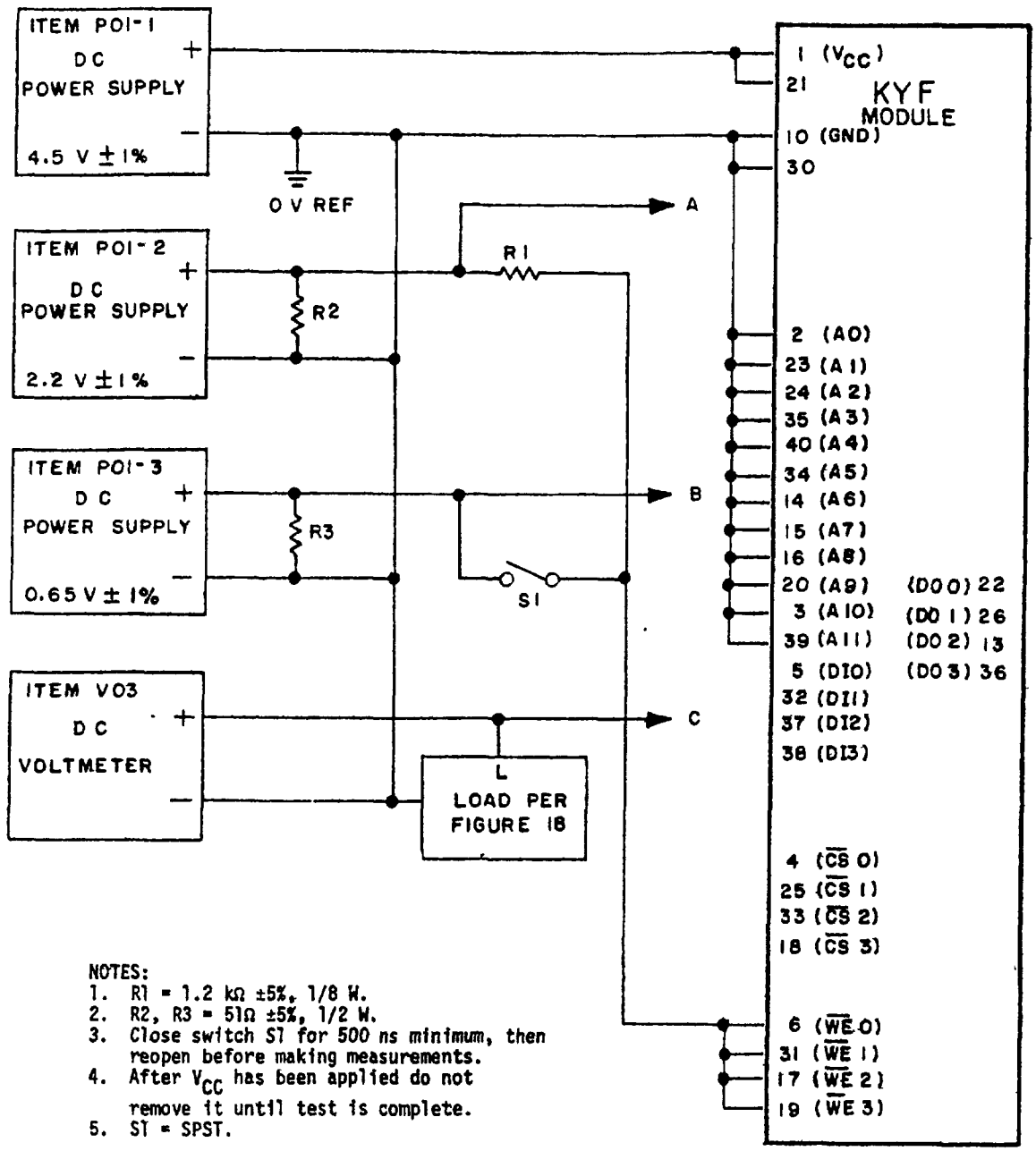
FIGURE 16. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Static output voltage (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	DO0 Static output voltage, low level	Open	4,5,18, 25,33	22	0.01	0.4	V	17
H2	DO1		4,18,25, 32,33	26				
H3	DO2		4,18,25, 33,37	13				
H4	DO3 Static output voltage, low level	Open	4,18,25, 33,38	36	0.01	0.4		
J1	DO0 Static output voltage, high level	5	4,18,25, 32,33,37, 38	22	2.4	4.5		
J2	DO1	32	4,5,18,25, 33,37,38	26				
J3	DO2	37	4,5,18,25, 32,33,38	13				
J4	DO3 Static output voltage, high level	38	4,5,18,25, 32,33,37	36	2.4	4.5	V	17

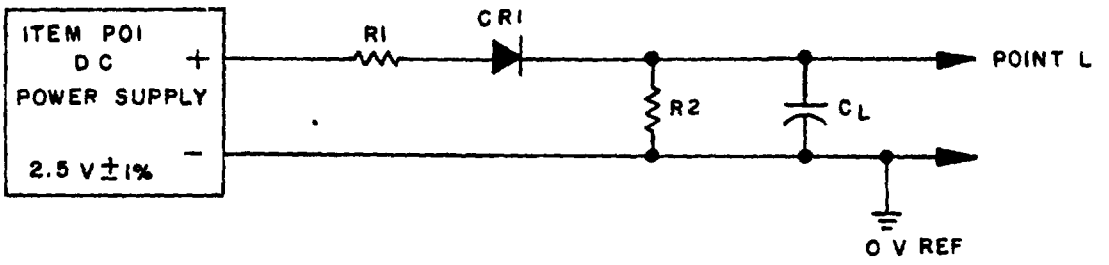
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NOTES:
1. R1 = 1.2 kΩ ±5%, 1/8 W.
2. R2, R3 = 51Ω ±5%, 1/2 W.
3. Close switch S1 for 500 ns minimum, then reopen before making measurements.
4. After V_{CC} has been applied do not remove it until test is complete.
5. S1 = SPST.

FIGURE 17. Static output voltage (low-level and high-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 1.2 kΩ ±1%, 1/8 W.
4. C_L < 10 pF (including probe and fixture capacitance) for test conditions R1-R8.
5. C_L = 50 ±5 pF (including probe and fixture capacitance) for test conditions Q1-Q4, Q9-Q12, S1-S6, and T1-T16.
6. C_L = 150 ±5 pF (including probe and fixture capacitance) for test conditions Q5-Q8, Q13-Q16, and S7-S12.
7. Capacitor C_L is not required for dc electrical test conditions H1-H4 and J1-J4.
8. Lead length from load circuit to module output pin under test is to be less than two inches (50.8 mm).

FIGURE 18. KYF logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	$\overline{CS0}$ Input current, high level	18,25,33	4	0	50	µA	19
K2	$\overline{CS1}$	4,18,33	25	↓	↓	↓	↓
K3	$\overline{CS2}$	4,18,25	33	↓	↓	↓	↓
K4	$\overline{CS3}$ Input current, high level	4,25,33	18	0	50	µA	19

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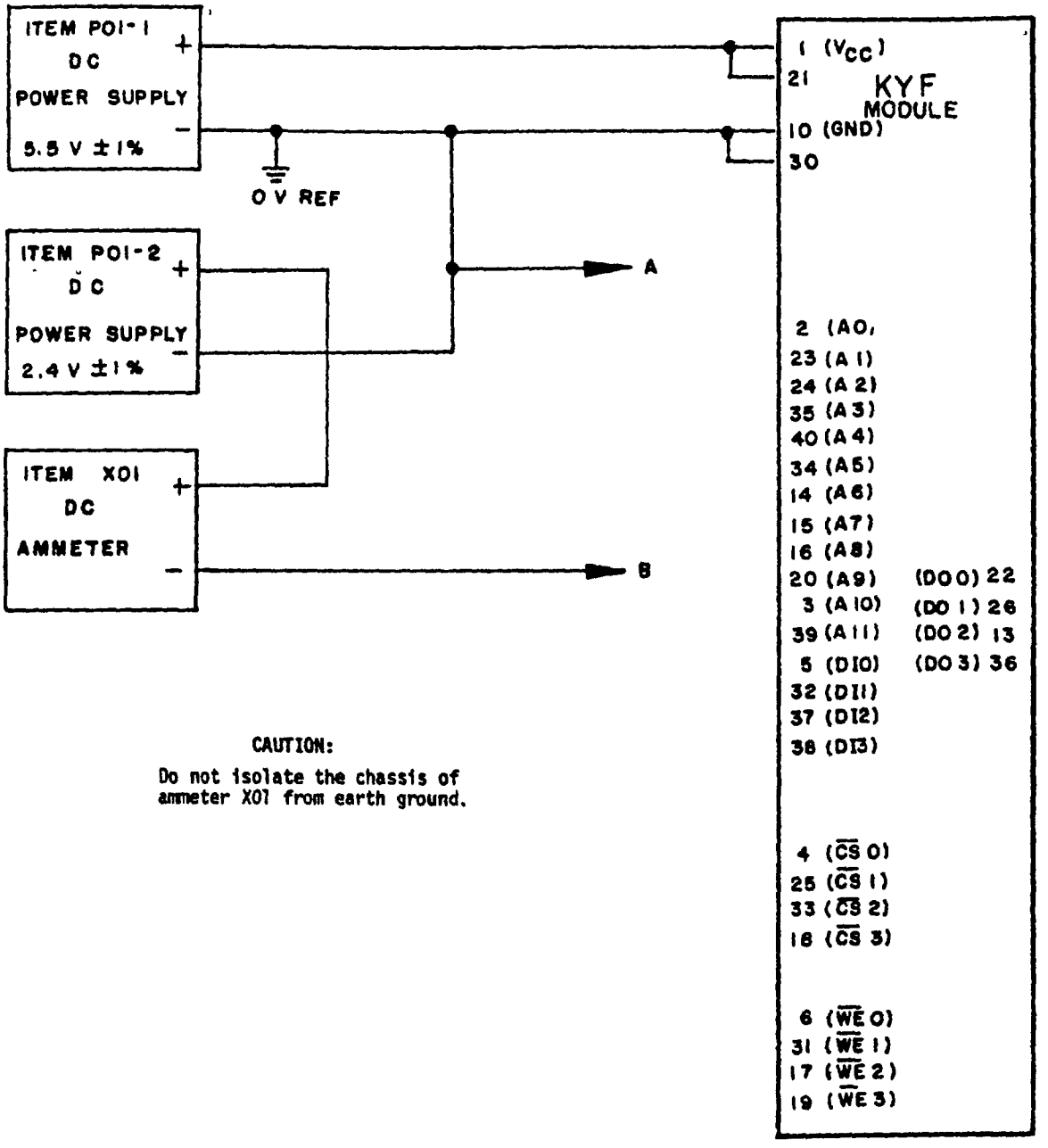


FIGURE 19. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	$\overline{CS0}$	Input current, low level	2,5,6,14,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37, 38,40,39	4	-0.1	-1.73	mA	20
L2	$\overline{CS1}$		2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,31,32 33,34,35,37,38 40,39	25				
L3	$\overline{CS2}$		2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,34,35,37,38, 40,39	33				
L4	$\overline{CS3}$		2,4,5,6,14,15, 16,17,3,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	18	-0.1	-1.73	mA	
L5	A0		4,5,6,14,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	2	-10	1	µA	
L6	A1	Input current, low level	2,4,5,6,14,15, 16,17,3,18,19, 20,24,25,31,32, 33,34,35,37,38, 40,39	23	-10	1	µA	20

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TABLE XI. Input current (low-level) test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L7	A2 Input current, low level	2,4,5,6,14,15, 16,17,3,18,19, 20,23,25,31,32 33,34,35,37, 38,40,39	24	-10	1	µA	20
L8	A3	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31 32,33,34,37,38 40,39	35				
L9	A4	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31 32,33,34,35,37 38,39	40				
L10	A5	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31 32,33,35,37,38 40,39	34				
L11	A6	2,4,5,6,15,16, 17,3,18,19,20, 23,24,25,31,32 33,34,35,37, 38,40,39	14				
L12	A7 Input current, low level	2,4,5,6,14,16, 17,3,18,19,20, 23,24,25,31,32 33,34,35,37,38 40,39	15	-10	1	µA	20

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TABLE XI. Input current (low-level) test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L13	A8 Input current, low level	2,4,5,6,14,15, 17,3,18,19,20, 23,24,25,31,32 33,34,35,37,38 40,39	16	-10	1	µA	20
L14	A9	2,4,5,6,14,15, 16,17,3,18,19, 23,24,25,31,32 33,34,35,37, 38,40,39	20				
L15	A10	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	3				
L16	A11	2,4,5,6,14,15, 16,17,18,19,20, 3,23,24,25,31, 32,33,34,35, 37,38,40	39				
L17	$\overline{WE0}$	2,4,5,14,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	6				
L18	$\overline{WE1}$ Input current, low level	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25, 32,33,34,35, 37,38,40,39	31	-10	1	µA	20

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TABLE XI. Input current (low-level) test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L19	$\overline{WE2}$ Input current, low level	2,4,5,6,14,15, 16,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	17	-10	1	µA	20
L20	$\overline{WE3}$	2,4,5,6,14,15, 16,17,3,18,20, 23,24,25,31,32, 33,34,35,37, 38,40,39	19				
L21	DIO	2,4,6,14,15,16, 17,3,18,19,20, 23,24,25,31,32, 33,34,35,37,38, 40,39	5				
L22	DI1	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 33,34,35,37, 38,40,39	32				
L23	DI2	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,35,38, 40,39	37				
L24	DI3 Input current, low level	2,4,5,6,14,15, 16,17,3,18,19, 20,23,24,25,31, 32,33,34,35,37, 40,39	38	-10	1	µA	20

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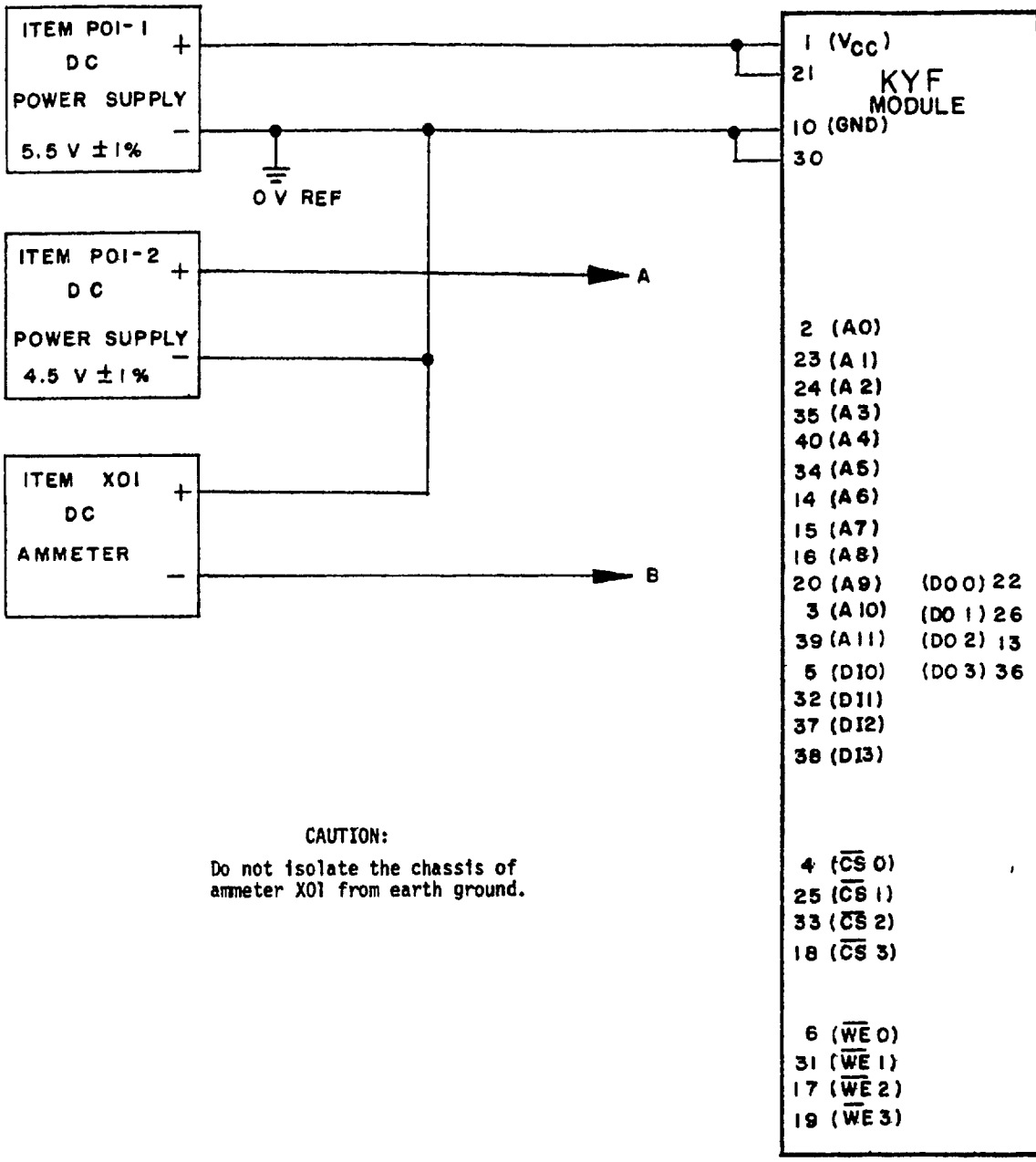


FIGURE 20. Input current (low-level) test circuit for table XI.

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TABLE XII. Address input threshold check (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Address threshold, low level	6,17,19, 31	2,14,15, 16,20,23, 24,34,35, 40,4,25, 18,33,3,39	22	0.0	0.4	V	21
M2				26				
M3				13				
M4	Address threshold, low level	6,17,19, 31	2,14,15, 16,20,23, 24,34,35, 40,4,25, 18,33,3,39	36	0.0	0.4		
M5	Address threshold, high level	2,6,14,15, 16,17,19, 20,23,24, 31,34,35, 40,3,39	4,18,25, 33	22	2.4	4.5		
M6				26				
M7				13				
M8	Address threshold, high level	2,6,14, 15,16,17, 19,20,23, 24,31,34, 35,40,3, 39	4,18,25, 33	36	2.4	4.5	V	21

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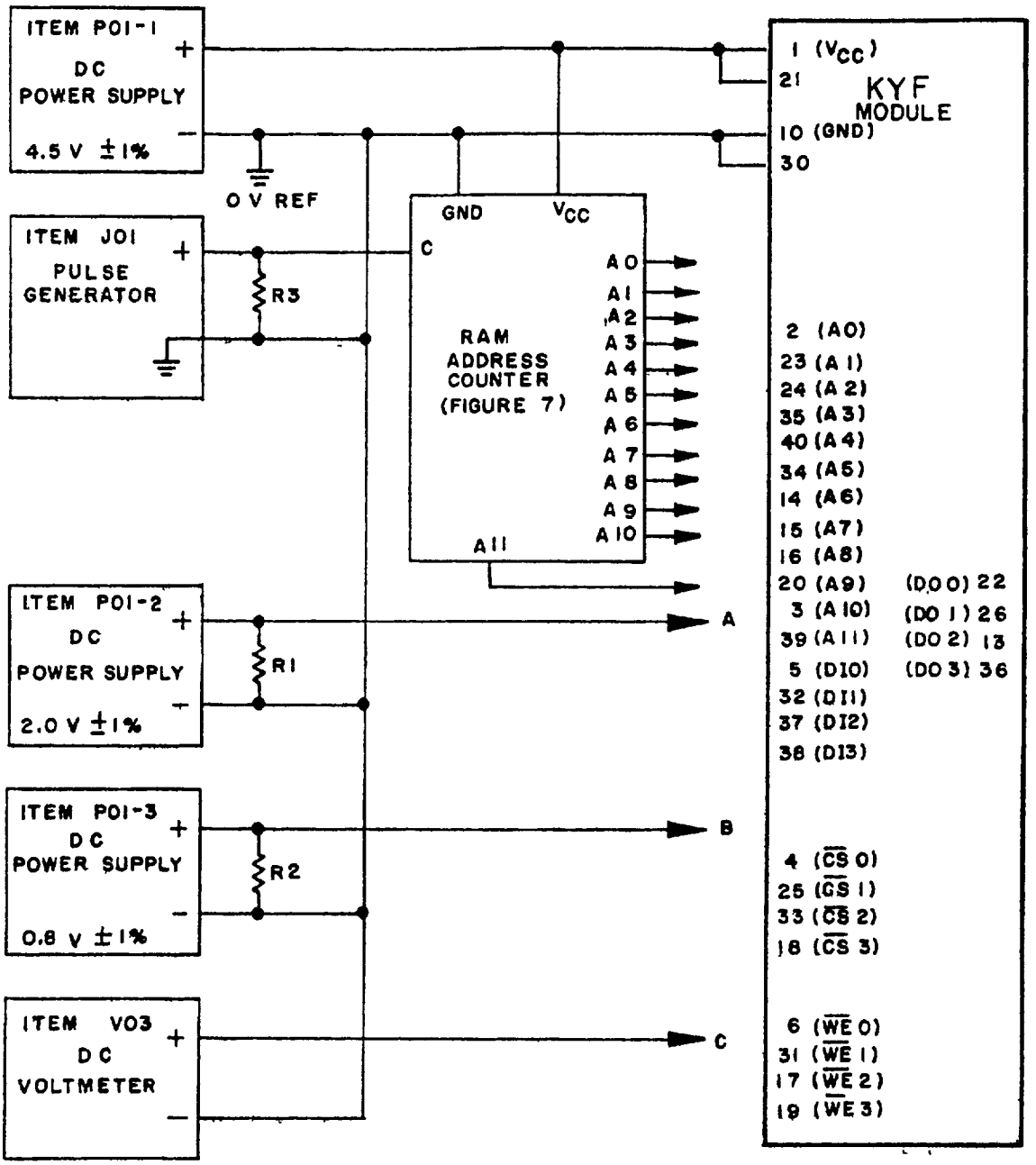


FIGURE 21. Address input threshold check (low-level and high-level) test circuit for table XII.

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NOTES:
1. R1, R2, R3 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 18.
3. Pulse generator (item J01) parameters: (See figure 25 for parameter definition.)

   a. Amplitude = 4.0 ±0.1 V.
   b. Pulse width = 1000 ±50 ns.
   c. Pulse repetition period = 2000 ns ±10%.
   d. t_{TLH}, t_{THL} = 25 ±20 ns.

4. Test sequence for test conditions M1-M4:

   a. Connect ram address counter to address inputs.
   b. Connect pins 5, 32, 37 and 38 to A. Connect pins 4, 6, 17, 18, 19, 25, 31 and 33 to B.
   c. Initiate ram address counter for five seconds. This loads all memory locations with a high level.
   d. Disconnect pins 6, 17, 19 and 31 from B and connect to A.
   e. Disconnect the ram address counter. Connect pins 2, 14, 15, 16, 20, 3, 39, 23, 24, 34, 35 and 40 to B. Connect pins 5, 32, 37 and 38 to B.
   f. Verify the outputs at location zero are high.
   g. Disconnect pins 6, 17, 19 and 31 from A and connect to B for a minimum of 500 ns, then connect to point A. Verify that this loads memory location zero with a low level and all other locations with a high level.
   h. Without disconnecting pins 6, 17, 19 and 31 from point A and without interrupting power perform tests as specified in test conditions M1-M4.

5. Test sequence for test conditions M5-M8:

   a. Connect ram address counter to address inputs.
   b. Connect pins 4, 5, 6, 17, 18, 19, 25, 31, 32, 33, 37 and 38 to B.
   c. Initiate ram address counter for five seconds. This loads all memory locations with a low level.
   d. Disconnect pins 6, 17, 19 and 31 from B and connect to A.
   e. Disconnect the ram address counter. Connect pins 2, 14, 15, 16, 20, 3, 39, 23, 24, 34, 35 and 40 to A. Connect pins 5, 32, 37 and 38 to A.
   f. Verify the outputs at location 4095 are low.
   g. Disconnect pins 6, 17, 19 and 31 from A and connect to B for a minimum of 500 ns, then connect to point A. Verify that this loads memory location 4095 with a high level and all other locations with a low level.
   h. Without disconnecting pins 6, 17, 19 and 31 from point A and without interrupting power perform tests as specified in test conditions M5-M8.

FIGURE 21. Address input threshold check (low-level and high-level) test circuit for table XII - Continued.

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TABLE XIII. Ground and V_{CC} pin continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	V_{CC} pin continuity	1	21	0.0	1.0	Ω	22
N2	Ground pin continuity	10	30		1.0		
N3	V_{CC} pin continuity	1	21		0.10		
N4	Ground pin continuity	10	30	0.0	0.10	Ω	22

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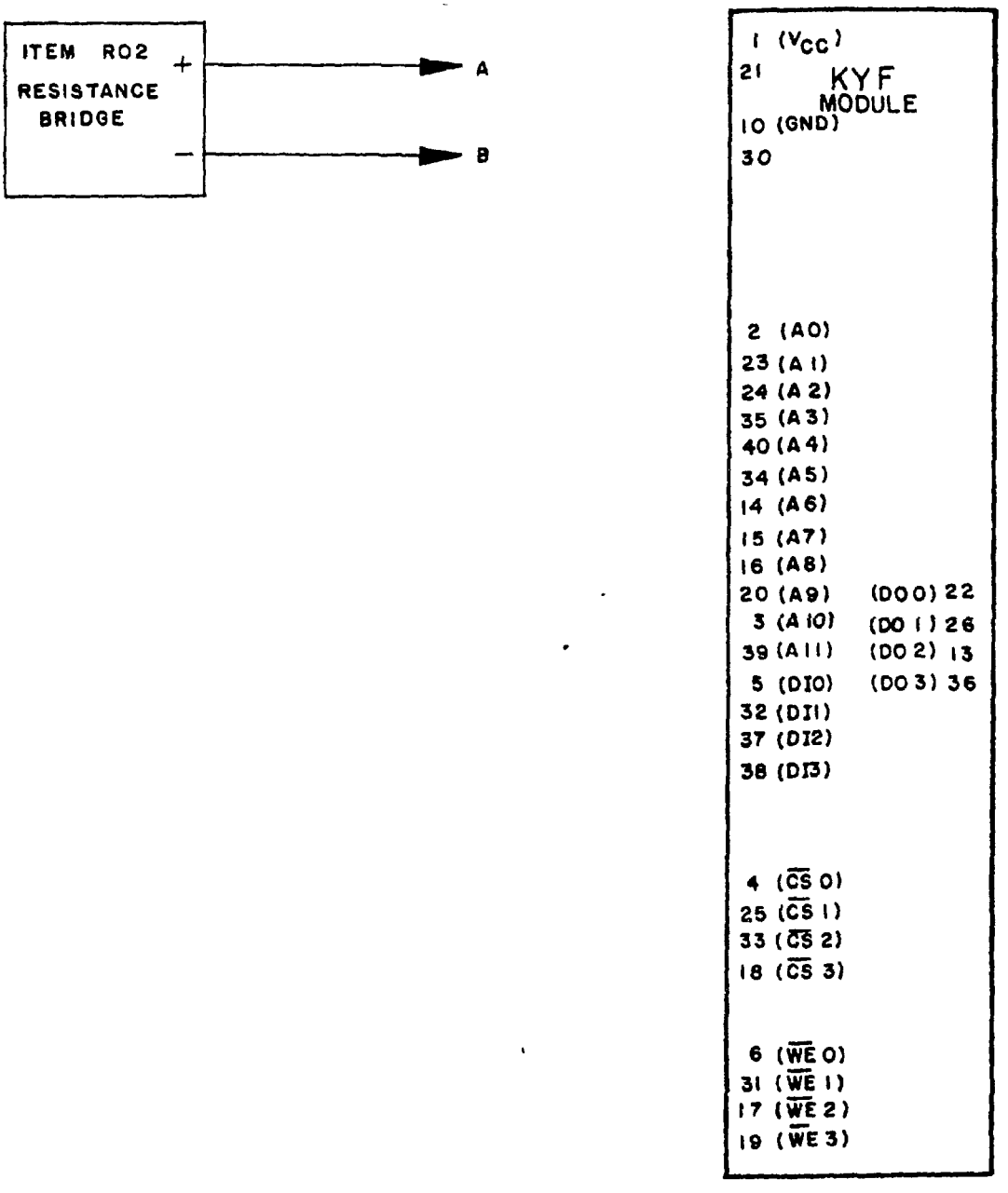


FIGURE 22. Ground and V_{CC} pin continuity test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply current	4,25,33,18	Open	50	694	mA	23
P2	↓	Open	4,25,33,18	↓	275	↓	↓
P3	↓	4	18,25,33	↓	↓	↓	↓
P4	↓	25	4,18,33	↓	↓	↓	↓
P5	↓	33	4,18,25	↓	↓	↓	↓
P6	Power supply current	18	4,25,33	50	275	mA	23

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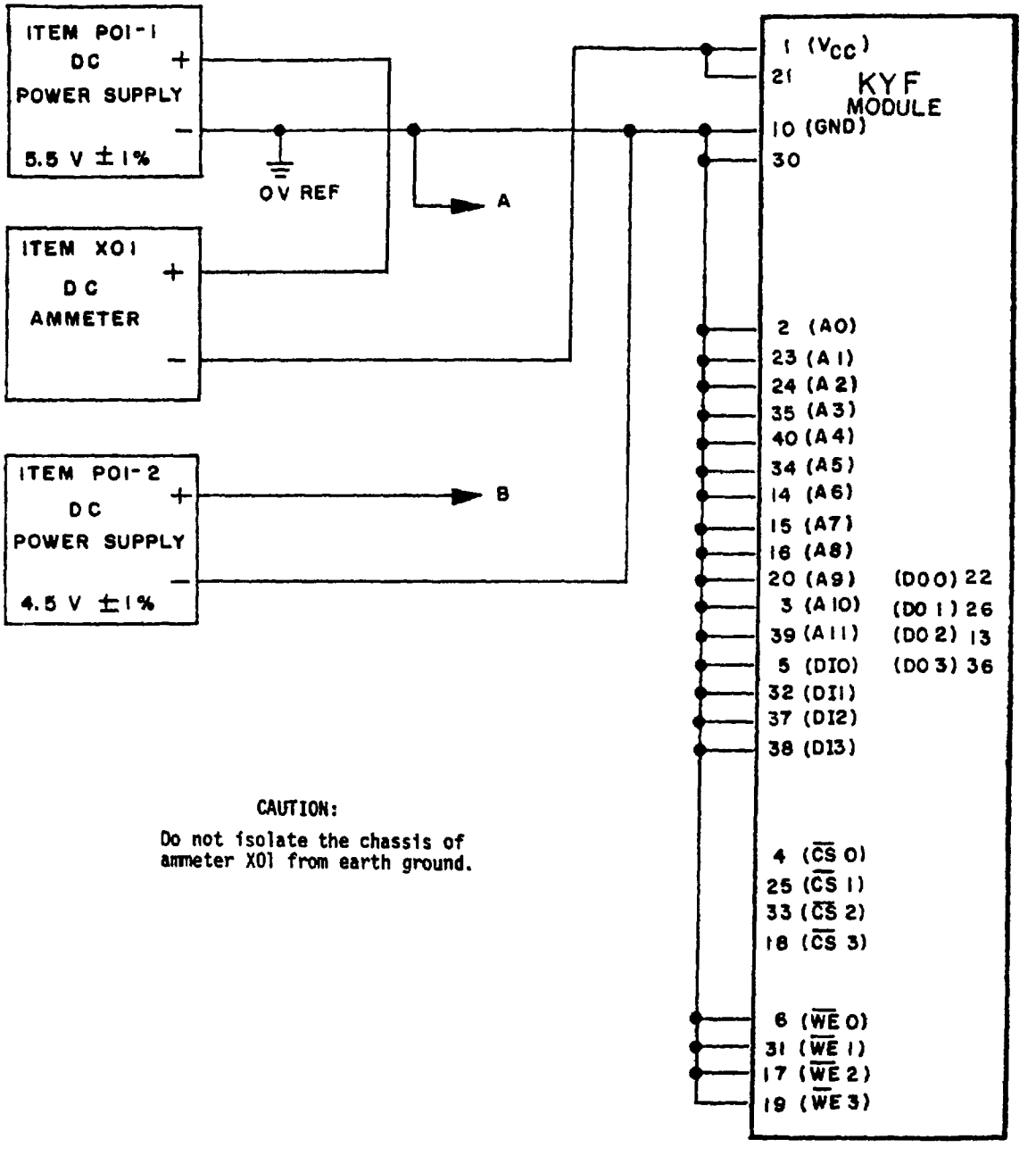


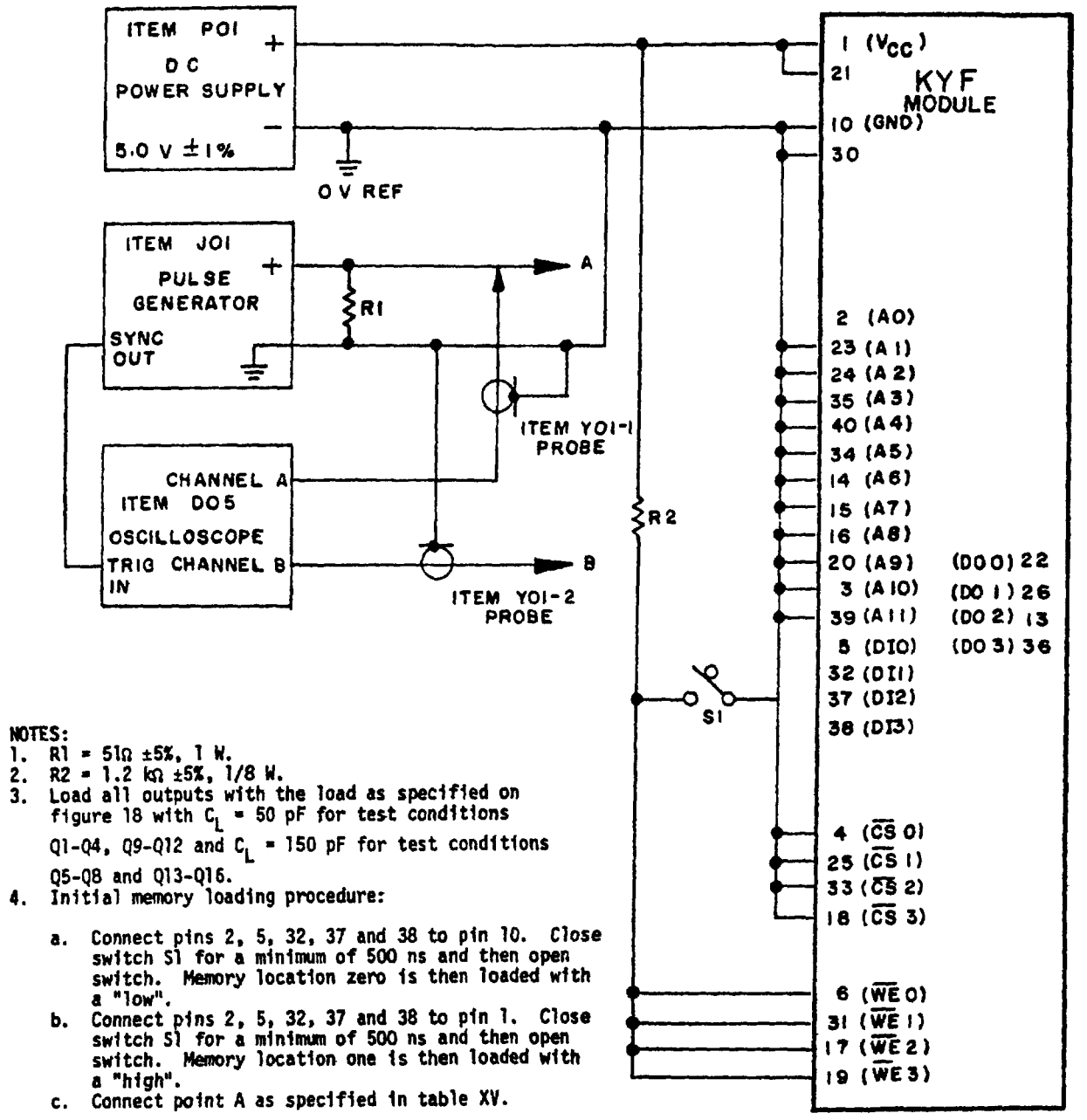
FIGURE 23. Power supply current test circuit for table XIV.

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TABLE XV. Transition time test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Q1	D00	Transition time (t_{TLH}), C_L = 50 pF	2	22	5	25	ns	24,25
Q2	D01			26				
Q3	D02			13				
Q4	D03	Transition time (t_{TLH}), C_L = 50 pF		36		25		
Q5	D00	Transition time (t_{TLH}), C_L = 150 pF		22		40		
Q6	D01			26				
Q7	D02			13				
Q8	D03	Transition time (t_{TLH}), C_L = 150 pF		36	5	40		
Q9	D00	Transition time (t_{THL}), C_L = 50 pF		22	3	15		
Q10	D01			26				
Q11	D02			13				
Q12	D03	Transition time (t_{THL}), C_L = 50 pF		36		15		
Q13	D00	Transition time (t_{THL}), C_L = 150 pF		22		25		
Q14	D01			26				
Q15	D02			13				
Q16	D03	Transition time (t_{THL}), C_L = 150 pF	2	36	3	25	ns	24,25

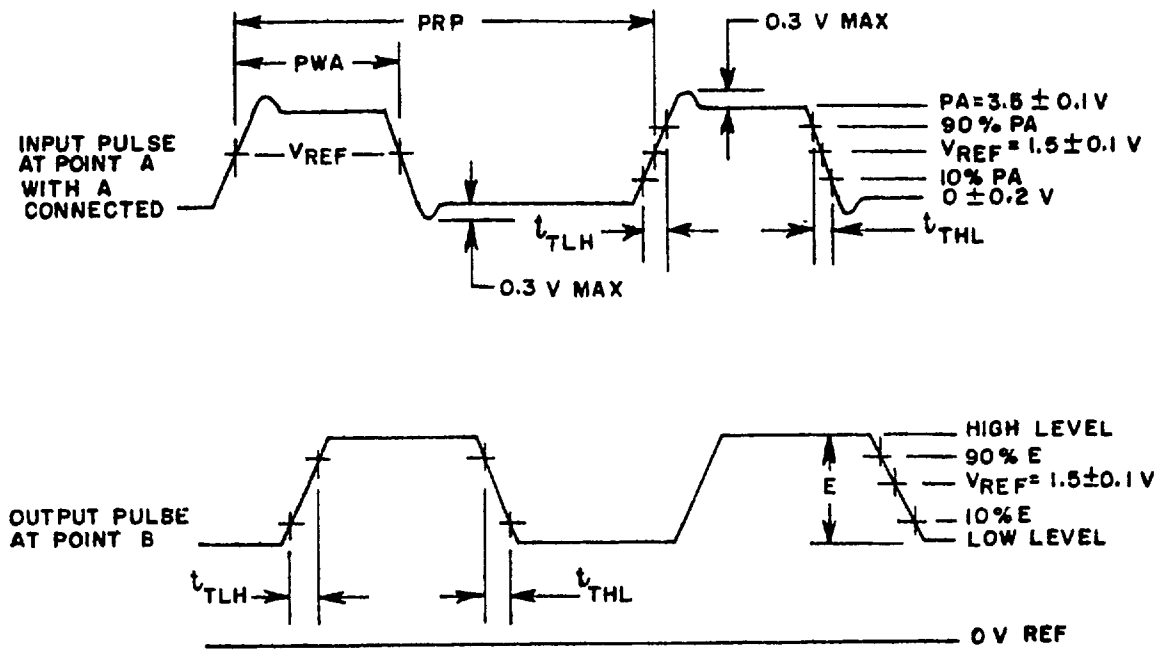
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1.2 kΩ ±5%, 1/8 W.
3. Load all outputs with the load as specified on figure 18 with C_L = 50 pF for test conditions Q1-Q4, Q9-Q12 and C_L = 150 pF for test conditions Q5-Q8 and Q13-Q16.
4. Initial memory loading procedure:
   a. Connect pins 2, 5, 32, 37 and 38 to pin 10. Close switch S1 for a minimum of 500 ns and then open switch. Memory location zero is then loaded with a "low".
   b. Connect pins 2, 5, 32, 37 and 38 to pin 1. Close switch S1 for a minimum of 500 ns and then open switch. Memory location one is then loaded with a "high".
   c. Connect point A as specified in table XV.
5. S1 = SPST.

FIGURE 24. Transition time test circuit for table XV.

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NOTES:
1. Input pulse parameters:

   t_{TLH}, t_{THL} = 10 ±2 ns.
   PWA = 1000 ±100 ns.
   PRP = 2000 ±200 ns.

2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.40 V.
3. All input voltages are referenced to ground.

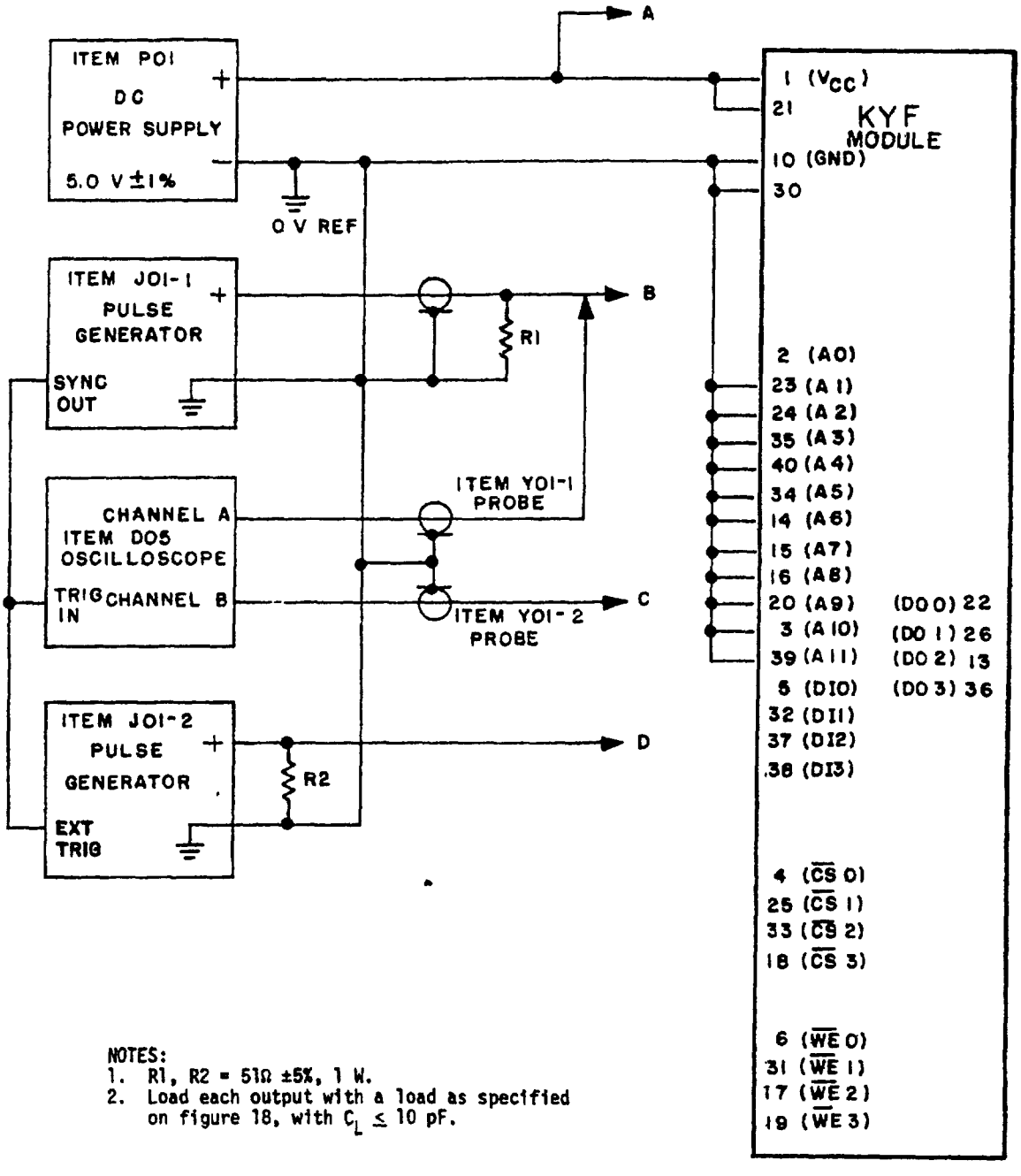
FIGURE 25. Transition time waveforms for table XV.

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TABLE XVI. Select recovery time test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	$\overline{CS0}$ Select recovery time (t_{SRL})	6,17,18 19,25, 31,33	4	22	2	5	40	ns	26,27
R2	$\overline{CS1}$	4,6,17, 18,19, 31,33	25	26					
R3	$\overline{CS2}$	4,6,17, 18,19, 25,31	33	13					
R4	$\overline{CS3}$ Select recovery time (t_{SRL})	4,6,17, 19,25, 31,33	18	36					
R5	$\overline{CS0}$ Select recovery time (t_{SRH})	6,17,18 19,25, 31,33	4	22					
R6	$\overline{CS1}$	4,6,17, 18,19, 31,33	25	26					
R7	$\overline{CS2}$	4,6,17, 18,19, 25,31	33	13					
R8	$\overline{CS3}$ Select recovery time (t_{SRH})	4,6,17, 19,25, 31,33	18	36	2	5	40	ns	26,27

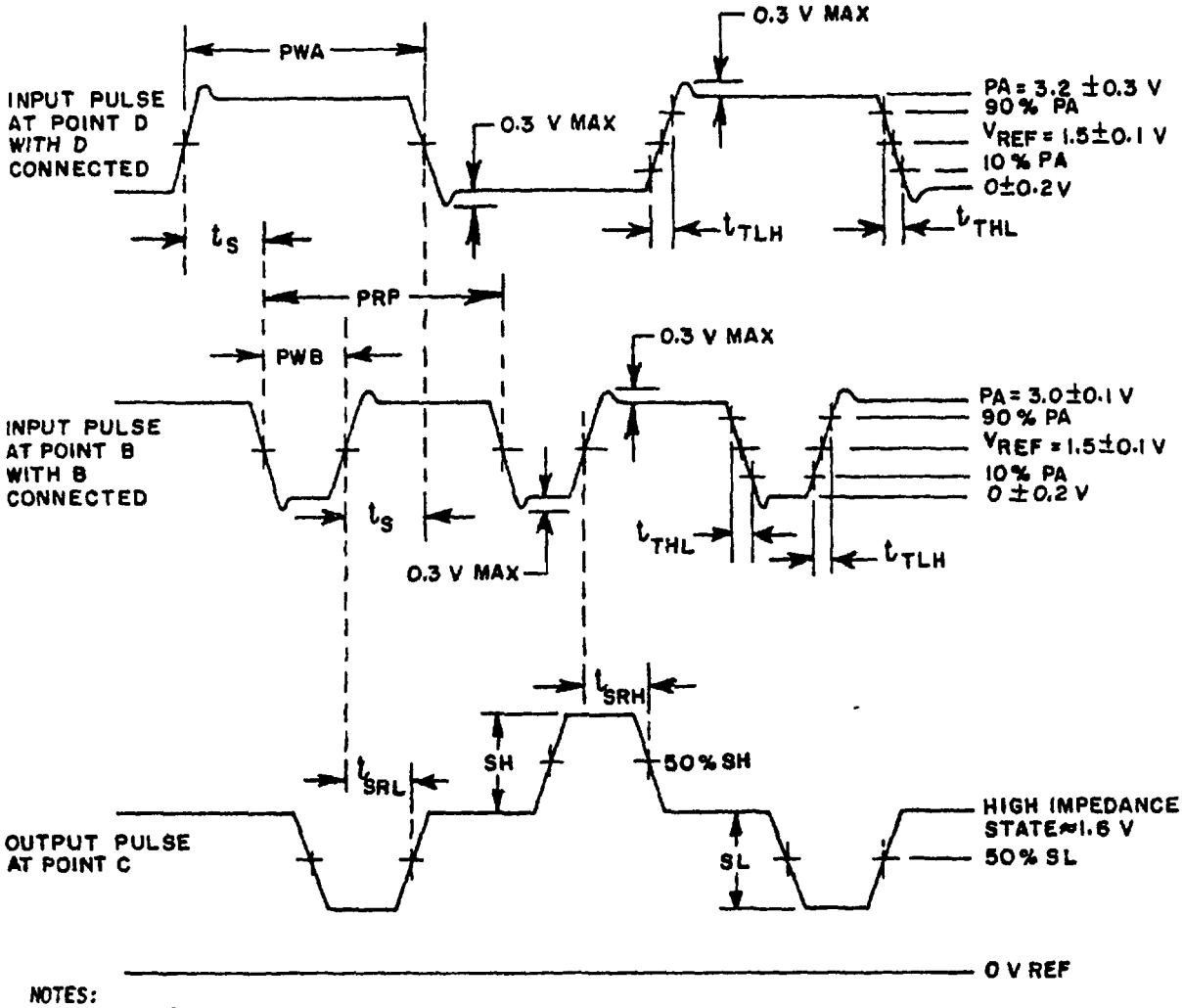
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NOTES:
1. R1, R2 = 51Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 18, with $C_L \leq 10$ pF.

FIGURE 26. Select recovery time test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 2000 ±100 ns.
   PWB = 600 ±50 ns.
   PRP = 2000 ±100 ns.
   t_S = 700 ±50 ns.
   t_{TLH}, t_{THL} = 10 ±2 ns.
2. Initial setup procedure:
   a. Store a "high" level in address location zero by connecting pins 2, 4, 18, 25, and 33 to pin 10 and connecting pins 5, 32, 37 and 38 to pin 1. Connect pins 6, 17, 19 and 31 to pin 10 for 400 ns minimum, then connect to point A.
   b. Store a "low" level in address location one by connecting pin 2 to pin 1 and connecting pins 4, 5, 18, 25, 32, 33, 37 and 38 to pin 10. Connect pins 6, 17, 19 and 31 to pin 10 for 400 ns minimum, then connect to point A.
   c. Verify that address location zero has a "high" stored and location one has a "low" stored.

FIGURE 27. Select recovery time waveforms for table XVI.

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TABLE XVII. Dynamic test.

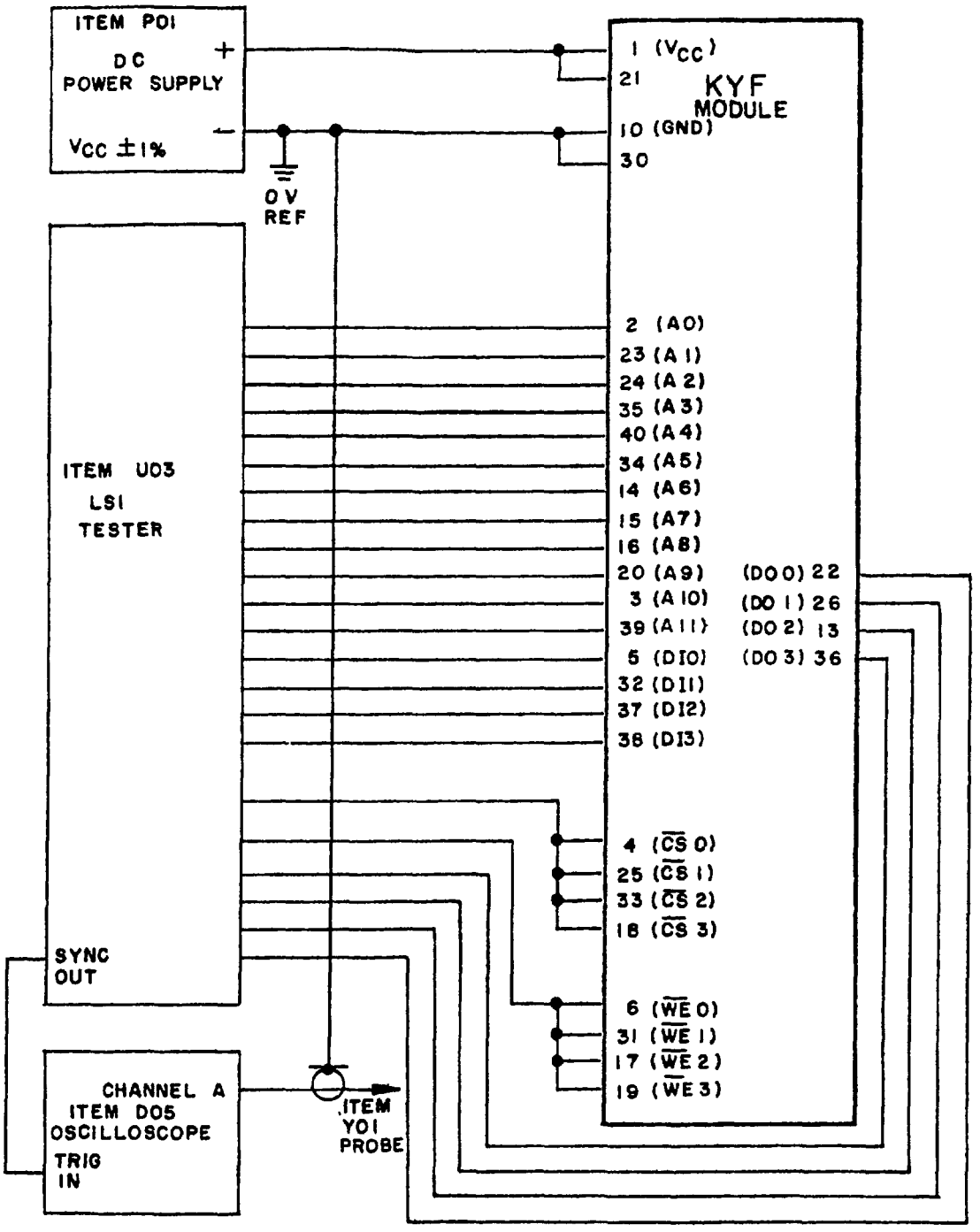
Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
S1	Inter-address write recovery-one, inter-address write recovery-zero pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 29 requirements			28,29
S2	Inter-address write recovery-one, inter-address write recovery-zero pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 29 requirements			28,29
S3	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 30 requirements			28,30
S4	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 30 requirements			28,30
S5	Static hold test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 31 requirements			28,31
S6	Static hold test pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 31 requirements			28,31
S7	Inter-address write recovery-one, inter-address write recovery-zero pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 29 requirements			28,29
S8	Inter-address write recovery-one, inter-address write recovery-zero pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 29 requirements			28,29
S9	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 30 requirements			28,30
S10	Marching write recovery-one, marching write recovery-zero, and circuit-select/deselect and chip independence pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 30 requirements			28,30

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TABLE XVII. Dynamic test -Continued.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
S11	Static hold test pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 31 requirements			28,31
S12	Static hold test pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 31 requirements			28,31

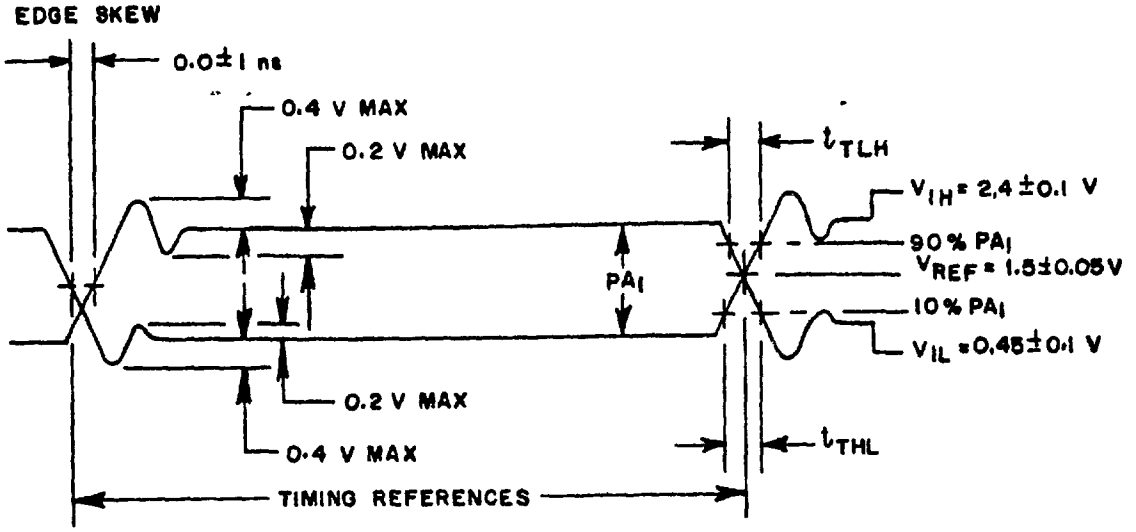
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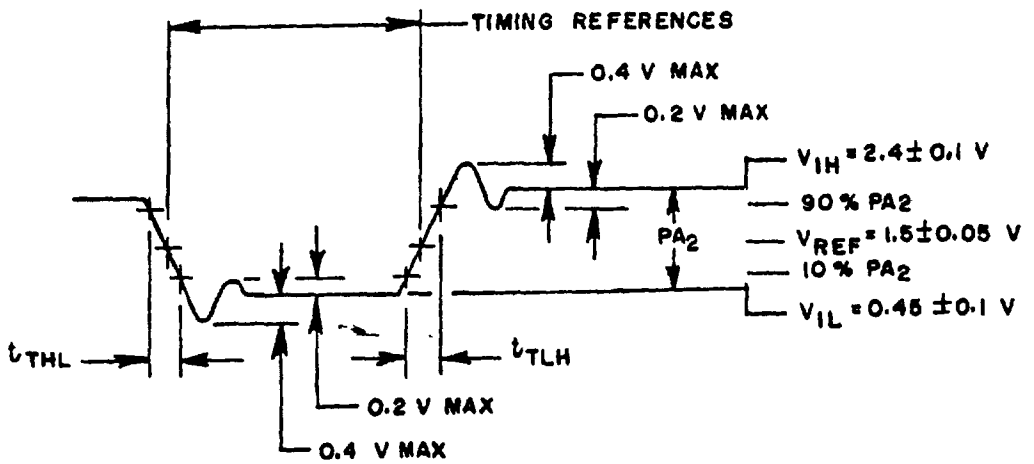
NOTES.
1. V_{CC} = 4.5 V ±1% for test conditions S1, S3, S5, S7, S9 and S11.
2. V_{CC} = 5.5 V ±1% for test conditions S2, S4, S6, S8, S10 and S12.
3 Load output pins 13, 22, 26 and 36 with load as specified on figure 18 with C_L = 50 pF for test conditions S1-S6.
4. Load output pins 13, 22, 26 and 36 with load as specified on figure 18 with C_L = 150 pF for test conditions S7-S12.
5. Reference document drawing number 3202946G1 for item U03/module interface definition.

FIGURE 28. Dynamic test circuit for table XVII.

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BIDIRECTIONAL INPUT WAVEFORMS

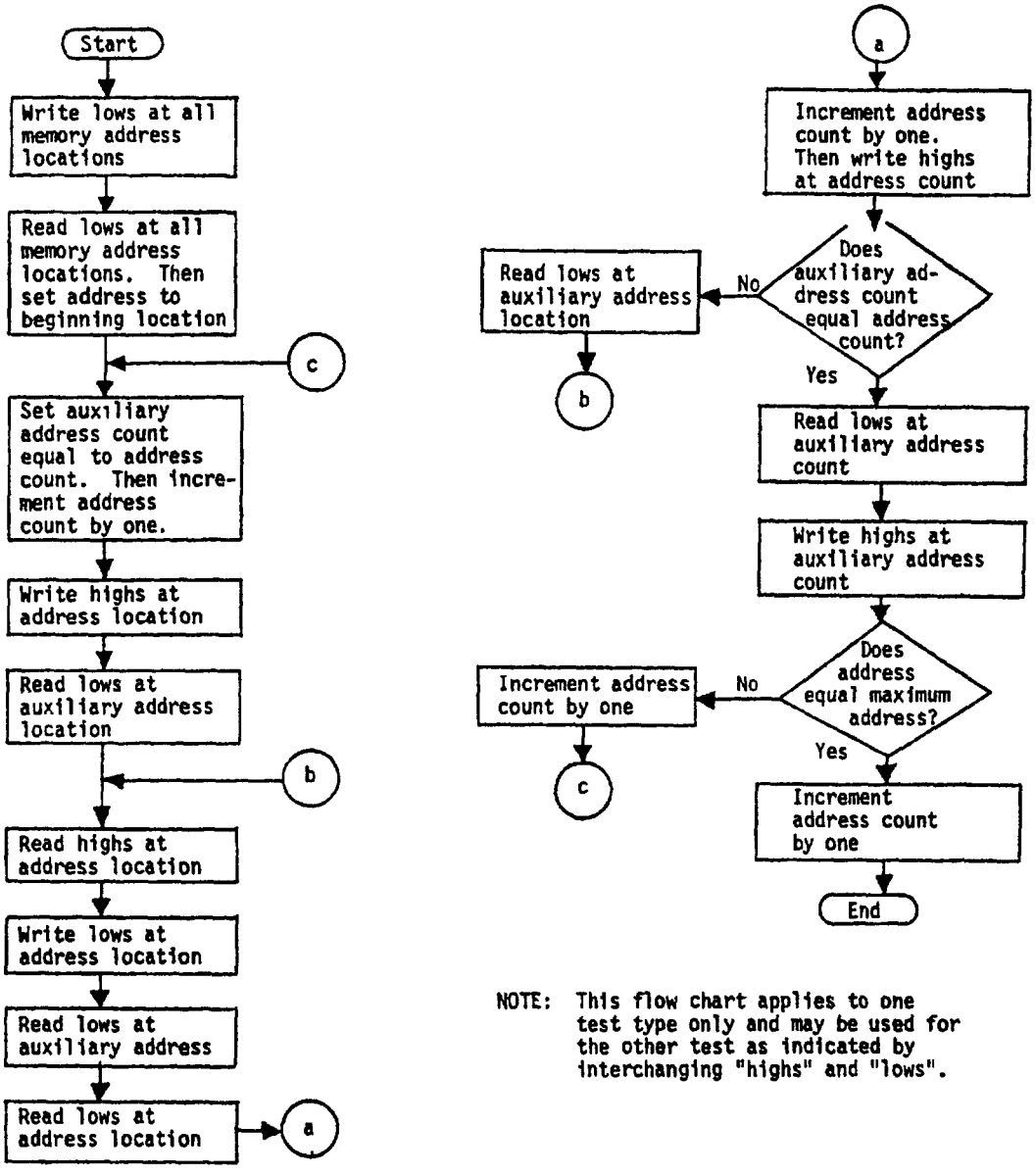


UNIDIRECTIONAL INPUT WAVEFORMS

NOTES:
1. t_{THL}, t_{TLH} = 10 ±4 ns.
2. All timing references are made at the V_{REF} level.

FIGURE 28. Dynamic test circuit input waveforms for table XVII.

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NOTE: This flow chart applies to one test type only and may be used for the other test as indicated by interchanging "highs" and "lows".

Program Flow Chart

FIGURE 29. Inter-address write recovery-one, inter-address write recovery-zero test pattern program flow chart, test waveforms, and program for table XVII.

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NOTES for program flow chart:

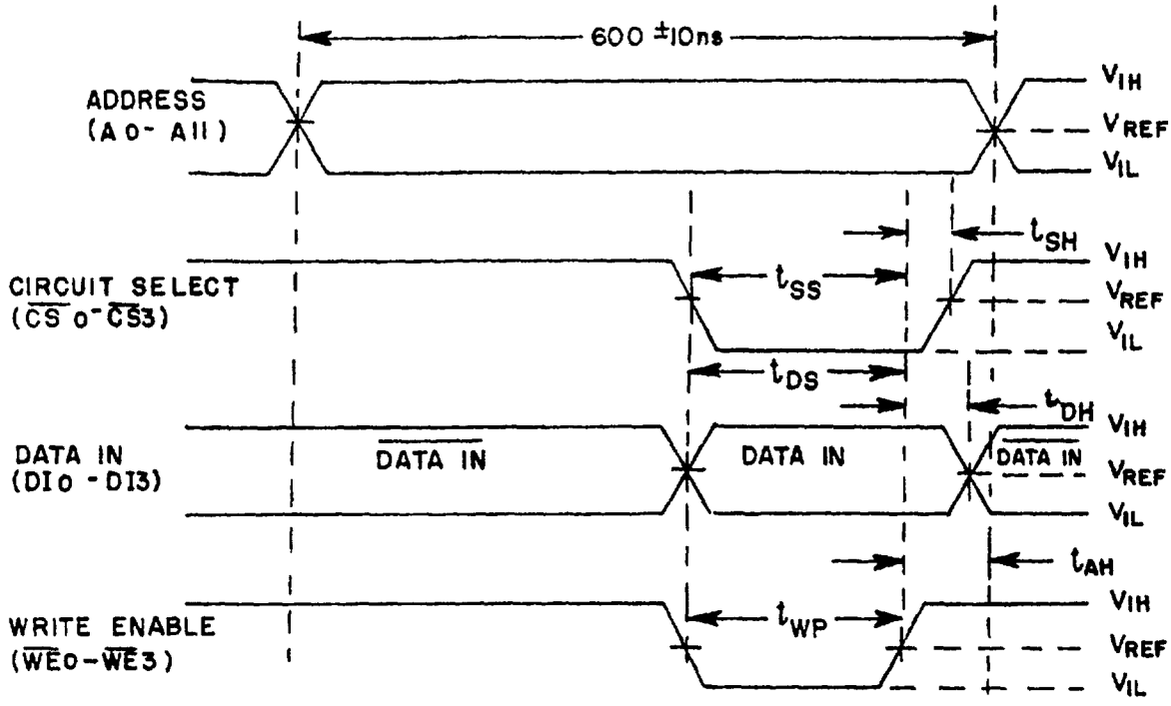
1. Inter-address write recovery-one, inter-address write recovery-zero pattern testing. The objective of the inter-address write recovery-one, inter-address write recovery-zero test pattern is to test the recovery time from a write at one address location to read at a different address location. It is necessary to check all possible address transitions from a write cycle at one address to a read cycle at a different address. This pattern also verifies address hold time (t_{AH}), minimum write pulse width (t_{WP}), address access time (t_{AA}), circuit select access time (t_{SA}), data setup time (t_{DS}), circuit select setup time (t_{SS}), and data hold time (t_{DH}).
2. A program that fulfills the inter-address write recovery-one test pattern requirements would:

   a. Write "lows" at all memory address locations.
   b. Read "lows" at all memory address locations.
   c. Write "highs" at memory address location one.
   d. Read "lows" at memory address location zero.
   e. Read "highs" at memory address location one.
   f. Write "lows" at memory address location one.
   g. Read "lows" at memory address location zero.
   h. Read "lows" at memory address location one.
   i. Continue the pattern until memory address location zero has been checked with respect to all other memory address locations.
   j. Write "highs" at memory address location two.
   k. Read "lows" at memory address location one.
   l. Read "highs" at memory address location two.
   m. Write "lows" at memory address location two.
   n. Read "lows" at memory address location one.
   o. Read "lows" at memory address location two.
   p. Write "highs" at memory address location zero.
   q. Read "lows" at memory address location one.
   r. Read "highs" at memory address location zero.
   s. Write "lows" at memory address location zero.
   t. Read "lows" at memory address location one.
   u. Read "lows" at memory address location zero.
   v. Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.

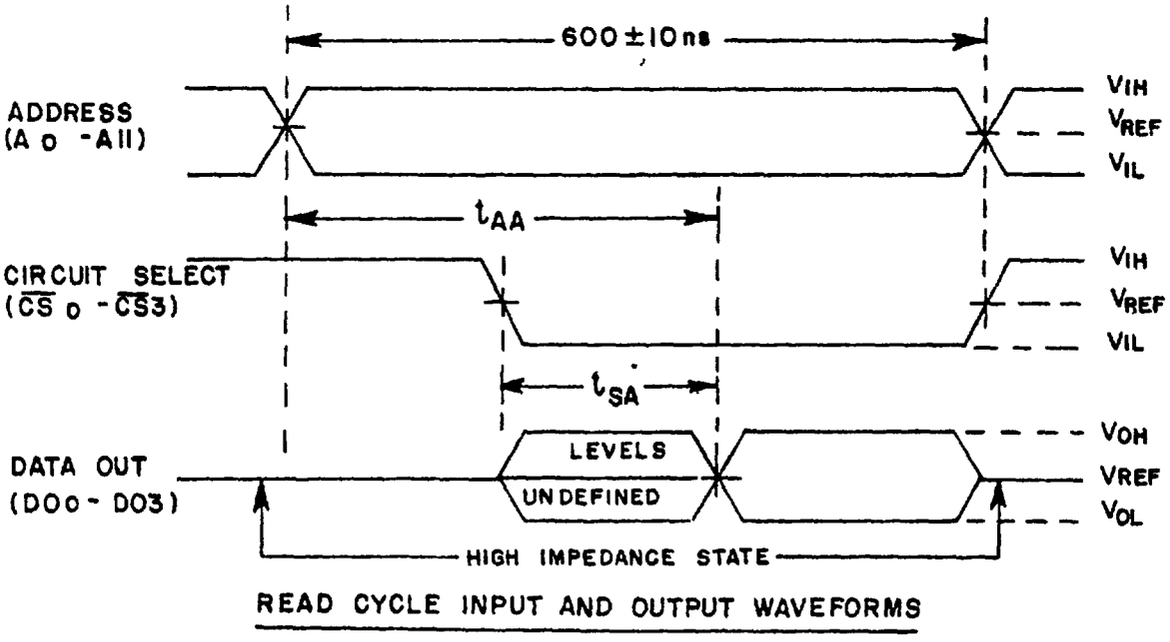
3. A program that fulfills the inter-address write recovery-zero test pattern requirements would be as in note 2, and figure 29 flow chart, with all "highs" and "lows" interchanged.
4. For test conditions S1 and S2, load each output pin with the load specified in figure 28, note 3.
5. For test conditions S7 and S8, load each output pin with the load specified in figure 28, note 4.
6. For test conditions S1 and S7, set V_{CC} = 4.5 V ±1%.
7. For test conditions S2 and S8, set V_{CC} = 5.5 V ±1%.

FIGURE 29. Inter-address write recovery-one, inter-address write recovery-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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WRITE CYCLE INPUT WAVEFORMS



READ CYCLE INPUT AND OUTPUT WAVEFORMS

FIGURE 29. Inter-address write recovery-one, inter-address write recovery-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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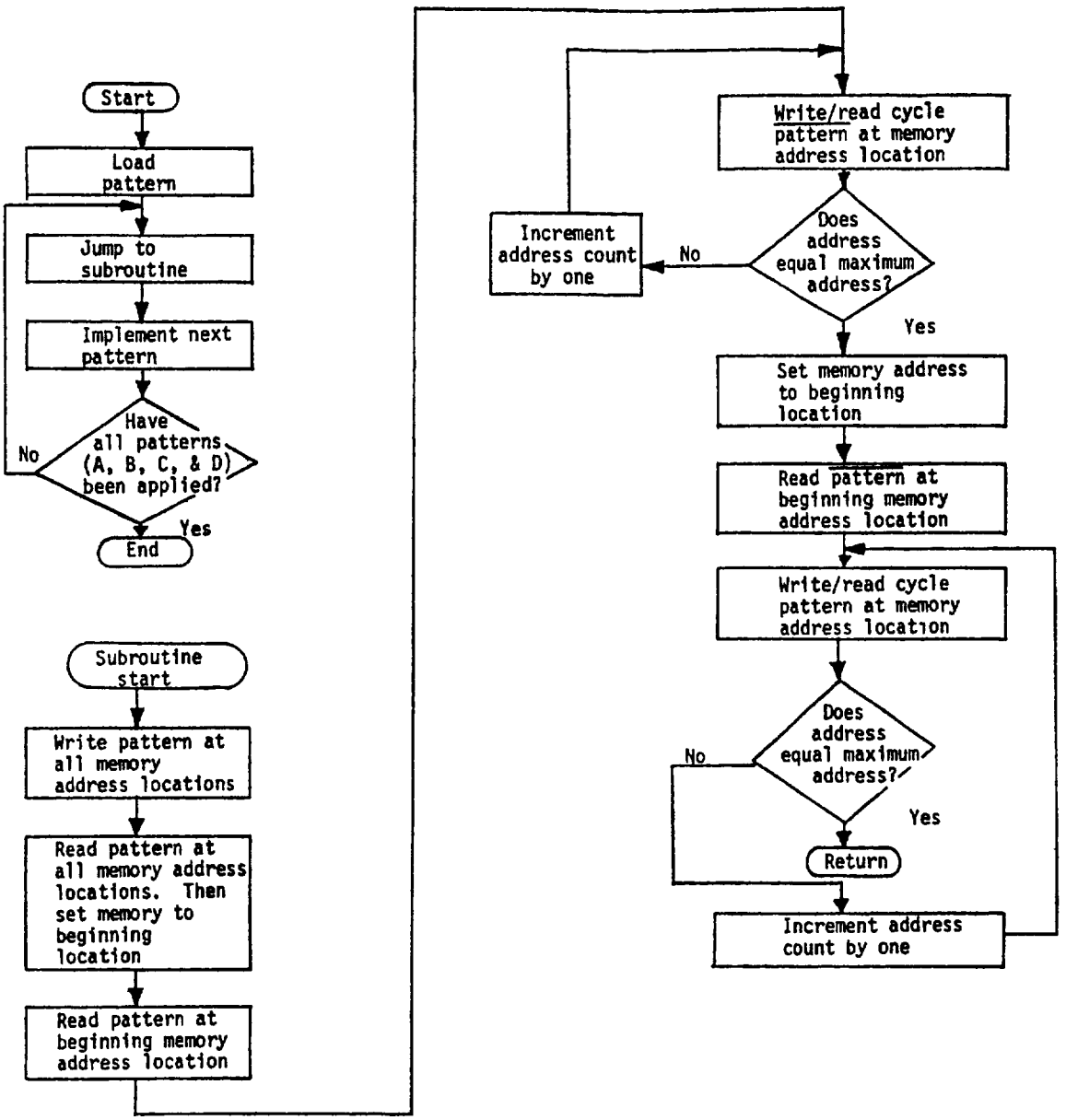
NOTES for write and read cycle waveforms:

1.	Data setup time (t_{DS})	300 +0 -5 ns
2.	Select setup time (t_{SS})	300 +0 -5 ns
3.	Address hold time (t_{AH})	100 +0 -5 ns
4.	Data hold time (t_{DH})	100 +0 -5 ns
5.	Select hold time (t_{SH})	50 +0 -5 ns
6.	Write pulse width (t_{WP})	300 +0 -5 ns
7.	Address access time (50 pF) (t_{AA})	525 +0 -5 ns
8.	Address access time (150 pF) (t_{AA})	530 +0 -5 ns
9.	Select access time (50 pF) (t_{SA})	375 +0 -5 ns
10.	Select access time (150 pF) (t_{SA})	380 +0 -5 ns
11.	Input high voltage (V_{IH})	2.2 ±0.1 V
12.	Input low voltage (V_{IL})	0.6 ±0.1 V
13.	Reference voltage (V_{REF})	1.5 ±0.1 V

14. All waveforms shall conform to the detailed input waveform requirements of figure 28.
15. All address lines need not switch simultaneously each cycle.

FIGURE 29. Inter-address write recovery-one, write recovery-zero test pattern program flow chart, test waveforms and program for table XVII - Continued.

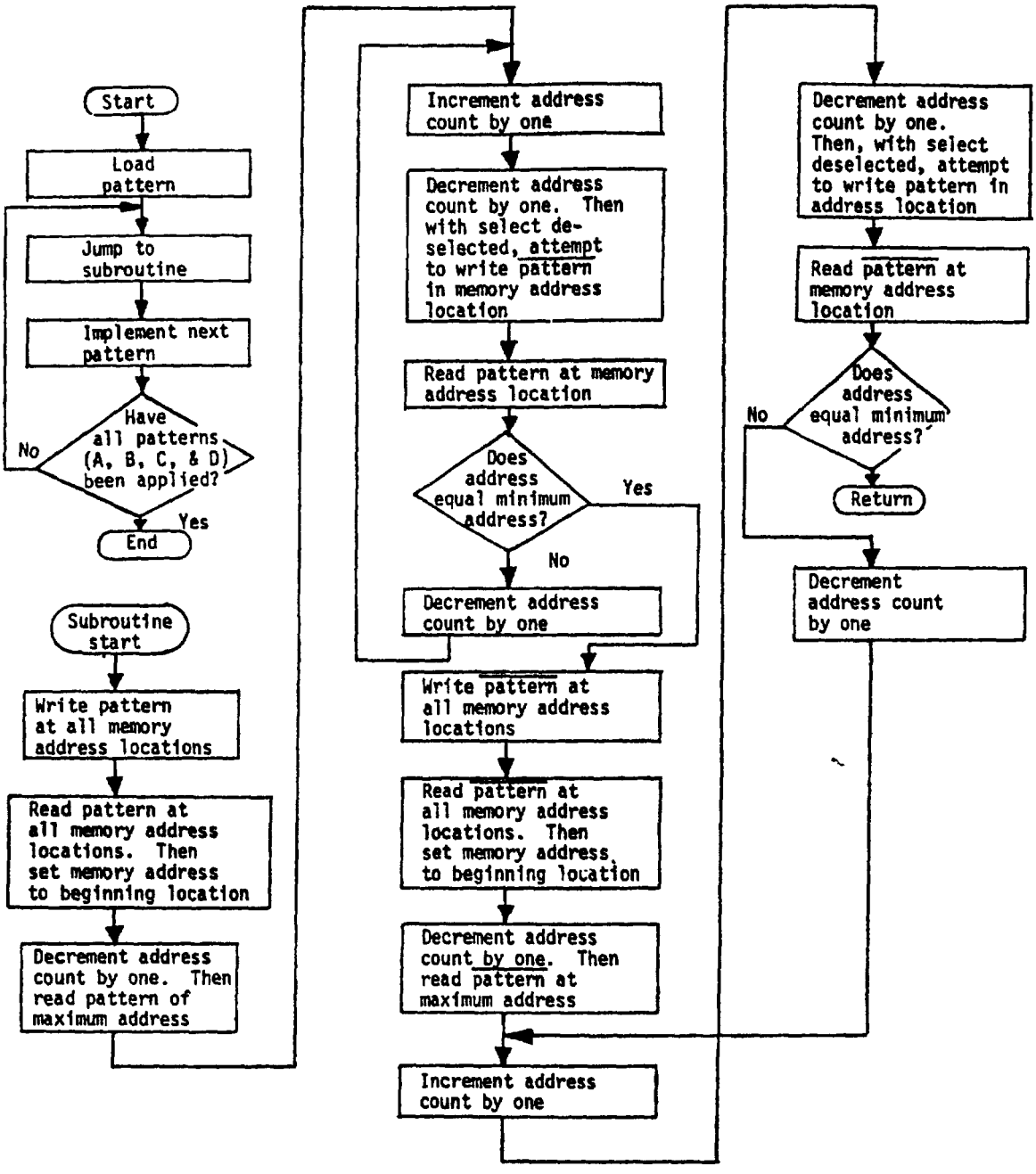
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:22Z
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Marching-write-recovery-one, marching-write-recovery-zero-pattern flow chart

FIGURE 30. Marching-write-recovery-one, marching-write-recovery-zero, and circuit-select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII.

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Circuit select/deselect pattern flow chart

NOTE: This pattern must be repeated four times using the data pattern specified in note 2.

FIGURE 30. Marching-write-recovery-one, marching-write-recovery-zero, and circuit-select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII.

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NOTES for marching write recovery-one and marching write recovery-zero pattern flow chart:

1. The objective of the marching write recovery-one, marching write recovery-zero test pattern is to check write recovery time of the module at the same address. The objective of the circuit select/deselect test pattern is to check for proper functioning of the address and circuit select circuitry in a dynamic environment. The patterns also verify address setup time (t_{AS}), circuit-select setup time (t_{SS}), circuit deselect setup time ($t_{\overline{SS}}$), circuit select hold time (t_{SH}), write recovery time (t_{WR}), write pulse width (t_{WP}), address access time (t_{AA}), select access time (t_{SA}), and data setup time (t_{DS}).
2. The chip independence test is a data pattern that is run during the marching write recovery test and the chip select/deselect test. This pattern checks that writing to one data bit does not affect other data bits. To run this pattern, the entire marching write recovery-one, marching write recovery-zero and chip select/deselect patterns are run on each of the following data patterns and their complements:

	DI3	DI2	DI1	DI0
A	0	0	0	1
B	0	0	1	0
C	0	1	0	0
D	1	0	0	0

3. A program that fulfills the marching write recovery-one, marching write recovery-zero test pattern requirements would:

   a. Write and read pattern A at all memory address locations.
   b. Read pattern A at address location zero.
   c. Write, then read pattern $\overline{A}$ at address location zero.
   d. Repeat (b) and (c) for address locations one through 4095, in turn.
   e. Set memory address location to zero.
   f. Read pattern $\overline{A}$ at address location zero.
   g. Write, then read pattern A at address location zero.
   h. Repeat (f) and (g) for address locations one through 4095, in turn.
   i. Repeat steps (a) through (h) with patterns B, C, and D.

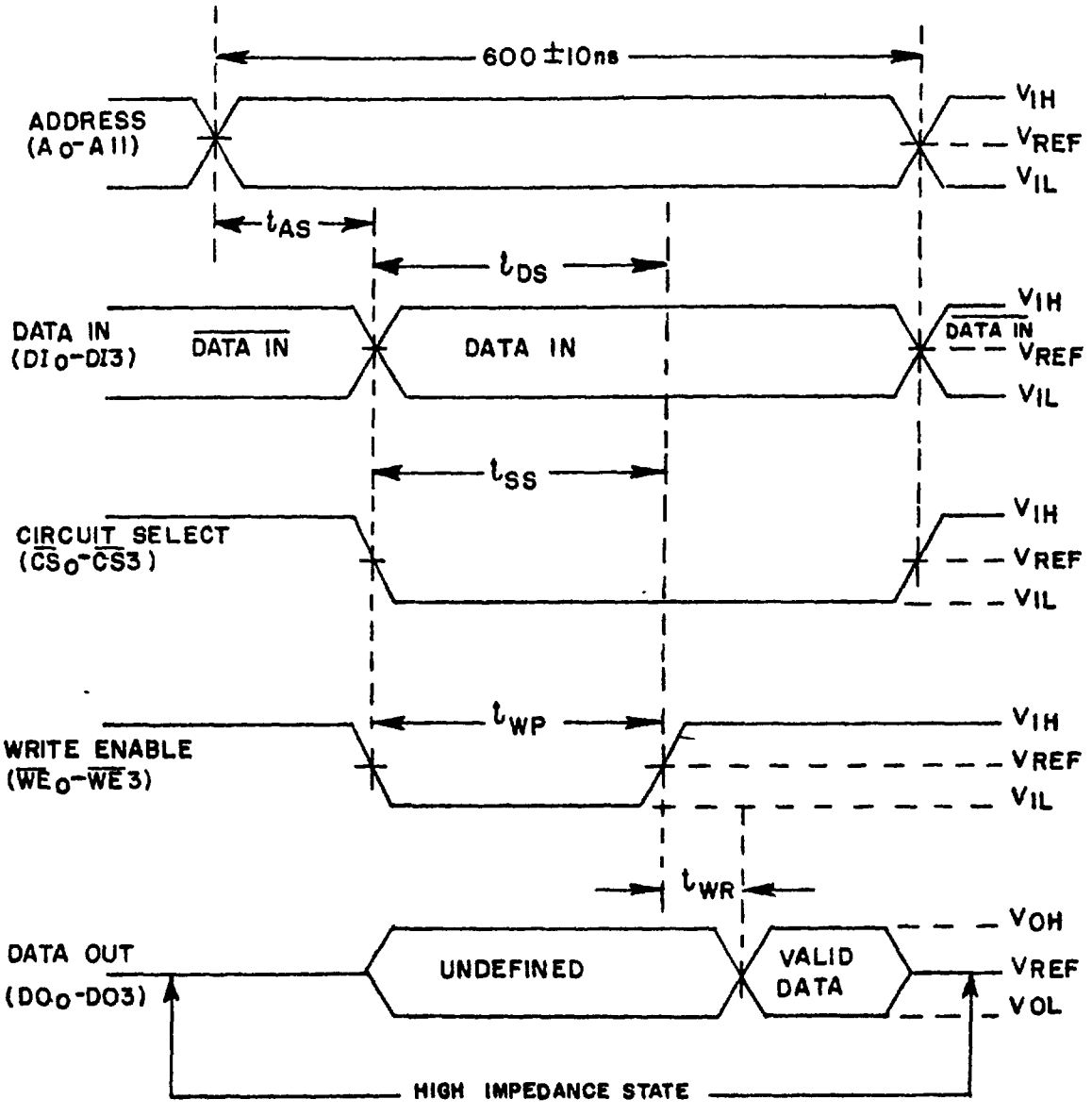
4. A program that fulfills the circuit-select/deselect test pattern requirements would:

   a. Write and read pattern A at all memory address locations.
   b. Set memory address to location zero.
   c. Decrement memory address by one, then read pattern A at maximum address.
   d. Increment memory address location by one.
   e. Decrement memory address location by one, then, with select deselected, attempt to write pattern $\overline{A}$ at the memory address location.
   f. Read pattern A at memory address location.
   g. Decrement address location by one and repeat (d) through (f) for the resultant address location.
   h. Repeat (g) until reading address location zero.
   i. Repeat steps (a) through (h) pattern A and pattern $\overline{A}$ interchanged.
   j. Repeat steps (a) through (i) with patterns B, C, and D.

5. For test conditions S3 and S4, load each output with the load specified in figure 28, note 3.
6. For test conditions S9 and S10, load each output with the load specified in figure 28, note 4.
7. For test conditions S3 and S9 set V_{CC} = 4.5 V ±1%.
8. For test conditions S4 and S10 set V_{CC} = 5.5 V ±1%.

FIGURE 30. Marching-write-recovery-one, marching-write-recovery-zero, and circuit-select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII. - Continued.

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WRITE READ CYCLE WAVEFORMS

FIGURE 30. Marching-write-recovery-one, marching-write-recovery-zero, and circuit-select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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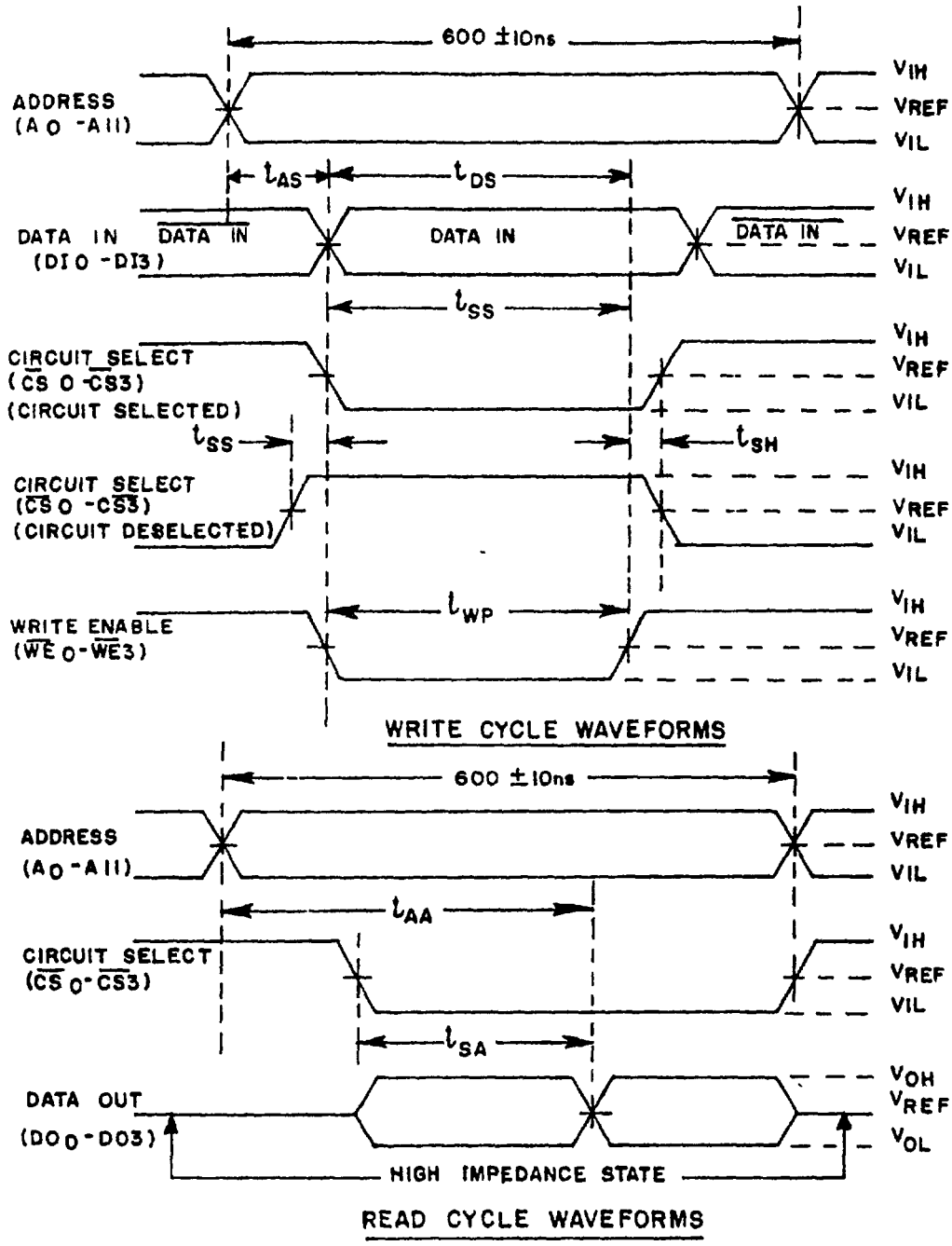


FIGURE 30. Marching-write-recovery-one, marking-write-recovery-zero, and circuit-select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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NOTES for write-read cycle waveforms:

1.	Address setup time (t_{AS})	150 +0 -5 ns
2.	Circuit-select setup time (t_{SS})	300 +0 -5 ns
3.	Circuit deselect time ($t_{\overline{SS}}$)	50 +3 -0 ns
4.	Circuit select hold time (t_{SH})	50 +3 -0 ns
5.	Data setup time (t_{DS})	300 +0 -5 ns
6.	Write pulse width (t_{WP})	300 +0 -5 ns
7.	Address access time, 50 pF load (t_{AA})	525 +0 -10 ns
8.	Address access time, 150 pF load (t_{AA})	530 +0 -10 ns
9.	Circuit-select access time, 50 pF load (t_{SA})	375 +0 -5 ns
10.	Circuit-select access time, 150 pF load (t_{SA})	380 +0 -5 ns
11.	Write recovery time, 50 pF load (t_{WR})	100 +0 -5 ns
12.	Write recovery time, 150 pF load (t_{WR})	110 +0 -5 ns
13.	All waveforms shall conform to detailed input waveform requirements of figure 28.	

FIGURE 30. Marching-write-recovery-one, marching-write-recovery-zero, and circuit-select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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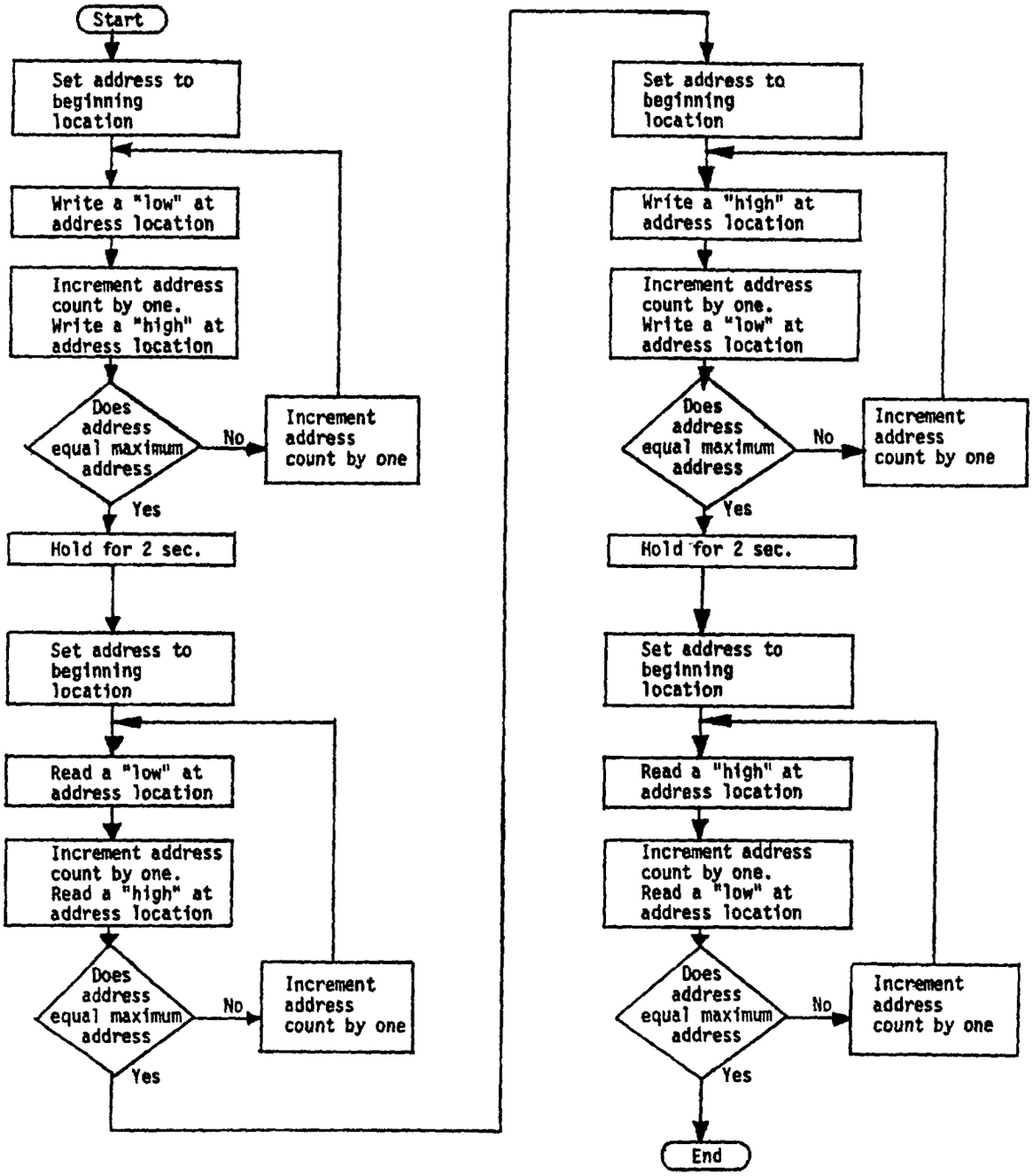


FIGURE 31. Static hold test pattern program flow chart, test waveforms, and program for table XVII.

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NOTES for static hold pattern program flow chart:

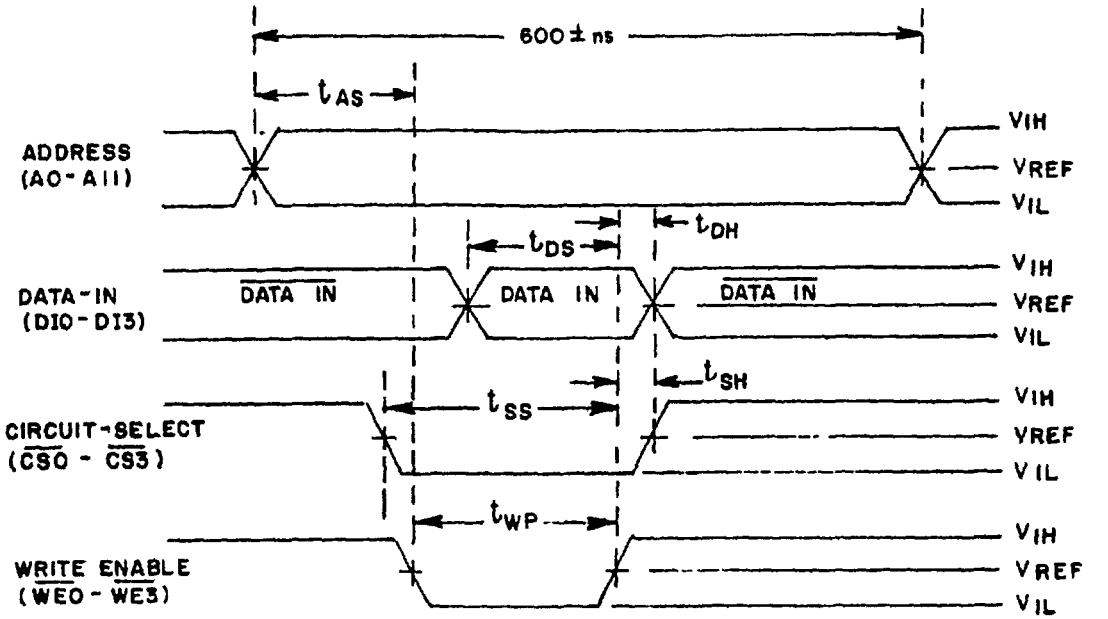
1. Static hold pattern testing: The objective of the static hold test pattern is to check data retention after a static hold period of 2 seconds. The pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), select setup time (t_{SS}), data setup time (t_{DS}), select hold time (t_{SH}), data hold time (t_{DH}), select access time (t_{SA}), and address access time (t_{AA}).
2. A pattern that fulfills the static hold test pattern requirements would:

   a. Write a "low" at address location zero.
   b. Write a "high" at address location one.
   c. Repeat (a) and (b) for address locations two through 4095.
   d. Hold the memory (no transitions) for 2 seconds.
   e. Read a "low" at address location zero.
   f. Read a "high" at address location one.
   g. Repeat (e) and (f) for address locations two through 4095.
   h. Write a "high" at address location zero.
   i. Write a "low" at address location one.
   j. Repeat (h) and (i) for address locations two through 4095.
   k. Hold the memory (no transitions) for 2 seconds.
   l. Read a "high" at address location zero.
   m. Read a "low" at address location one.
   n. Repeat (l) and (m) for address locations two through 4095.

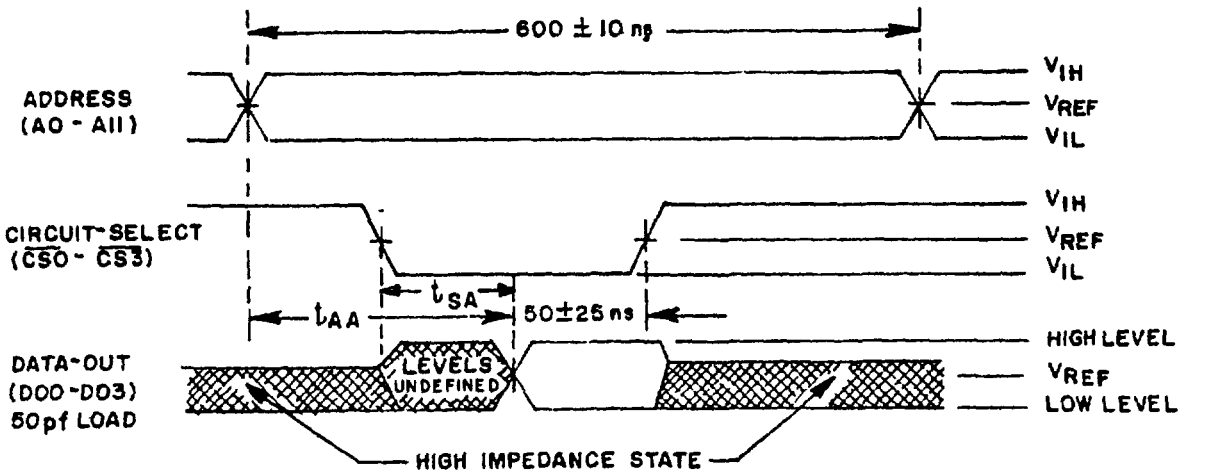
3. For test conditions S5 and S6, load each output with the load specified on figure 28, note 3.
4. For test conditions S11 and S12, load each output with the load specified in figure 28, note 4.
5. For test conditions S5 and S11, set V_{CC} = 4.5 V ±1%.
6. For test conditions S6 and S12, set V_{CC} = 5.5 V ±1%.

FIGURE 31. Static hold test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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WRITE CYCLE INPUT WAVEFORMS



READ CYCLE INPUT AND OUTPUT WAVEFORMS

FIGURE 31. Static hold test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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NOTES for write and read cycle input and output waveforms:

1.	Address setup time (t_{AS})	150 +0 -5 ns
2.	Data setup time (t_{DS})	300 +0 -5 ns
3.	Select setup time (t_{SS})	300 +0 -5 ns
4.	Data hold time (t_{DH})	100 +0 -5 ns
5.	Select hold time (t_{SH})	50 +0 -5 ns
6.	Write pulse width (t_{WP})	300 +0 -5 ns
7.	Address access time (50 pF) (t_{AA})	525 +0 -5 ns
8.	Address access time (150 pF) (t_{AA})	530 +0 -5 ns
9.	Select access time (50 pF) (t_{SA})	375 +0 -5 ns
10.	Select access time (150 pF) (t_{SA})	380 +0 -5 ns
11.	Input high voltage (V_{IH})	2.2 ±0.1 V
12.	Input low voltage (V_{IL})	0.6 ±0.1 V
13.	Reference voltage (V_{REF})	1.5 ±0.1 V

14. All waveforms shall conform to the detailed input waveform requirements of figure 28.
15. All address lines need not switch simultaneously each cycle.

FIGURE 31. Static hold test pattern program flow chart, test waveforms and program for table XVII - Continued.

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TABLE XVIII. Input crosstalk (low-level and high-level) test.

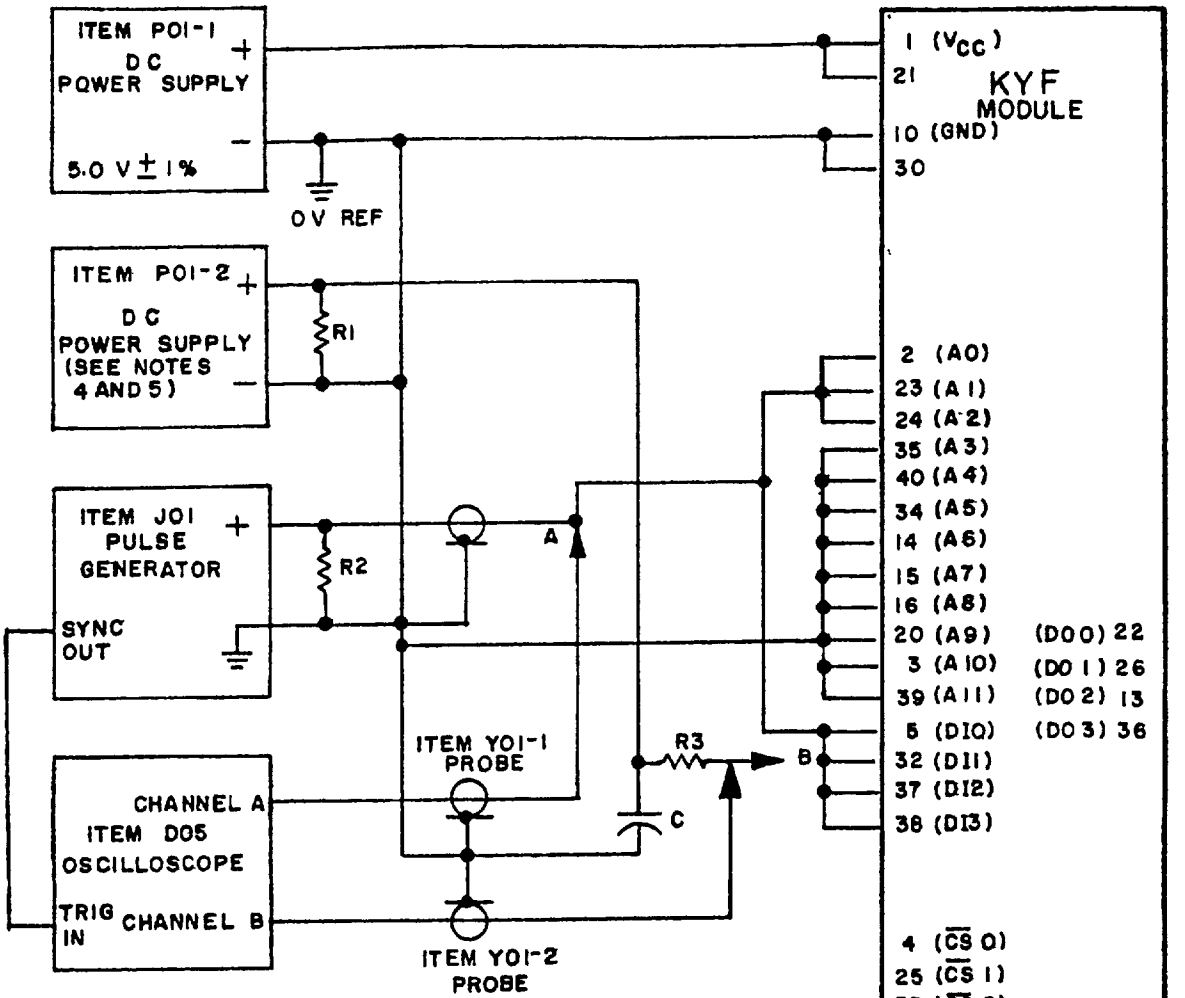
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
T1	$\overline{CS}0$	Input crosstalk, low level	6,17,18,19,25, 31,33	4	0	900	mV	32,33
T2	$\overline{CS}1$		4,6,17,18,19, 31,33	25				
T3	$\overline{CS}2$		4,6,17,18,19, 25,31	33				
T4	$\overline{CS}3$		4,6,17,19,25, 31,33	18		900		
T5	$\overline{WE}0$		4,17,18,19,25, 31,33	6		700		
T6	$\overline{WE}1$		4,6,17,18,19, 25,33	31				
T7	$\overline{WE}2$		4,6,18,19,25, 31,33	17				
T8	$\overline{WE}3$	Input crosstalk, low level	4,6,17,18,25, 31,33	19				
T9	$\overline{CS}0$	Input crosstalk, high level	6,17,18,19,25, 31,33	4				
T10	$\overline{CS}1$		4,6,17,18,19, 31,33	25				
T11	$\overline{CS}2$		4,6,17,18,19, 25,31	33				
T12	$\overline{CS}3$		4,6,17,19,25, 31,33	18		700		
T13	$\overline{WE}0$	Input crosstalk, high level	4,17,18,19,25, 31,33	6	0	500	mV	32,33

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TABLE XVIII. Input crosstalk (low-level and high-level) test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T14	$\overline{WE1}$ Input crosstalk, high level	4,6,17,18,19, 25,33	31	0	500	mV	32,33
T15	$\overline{WE2}$ Input crosstalk, high level	4,6,18,19,25, 31,33	17	0	500	mV	32,33
T16	$\overline{WE3}$ Input crosstalk, high level	4,6,17,18,25, 31,33	19	0	500	mV	32,33

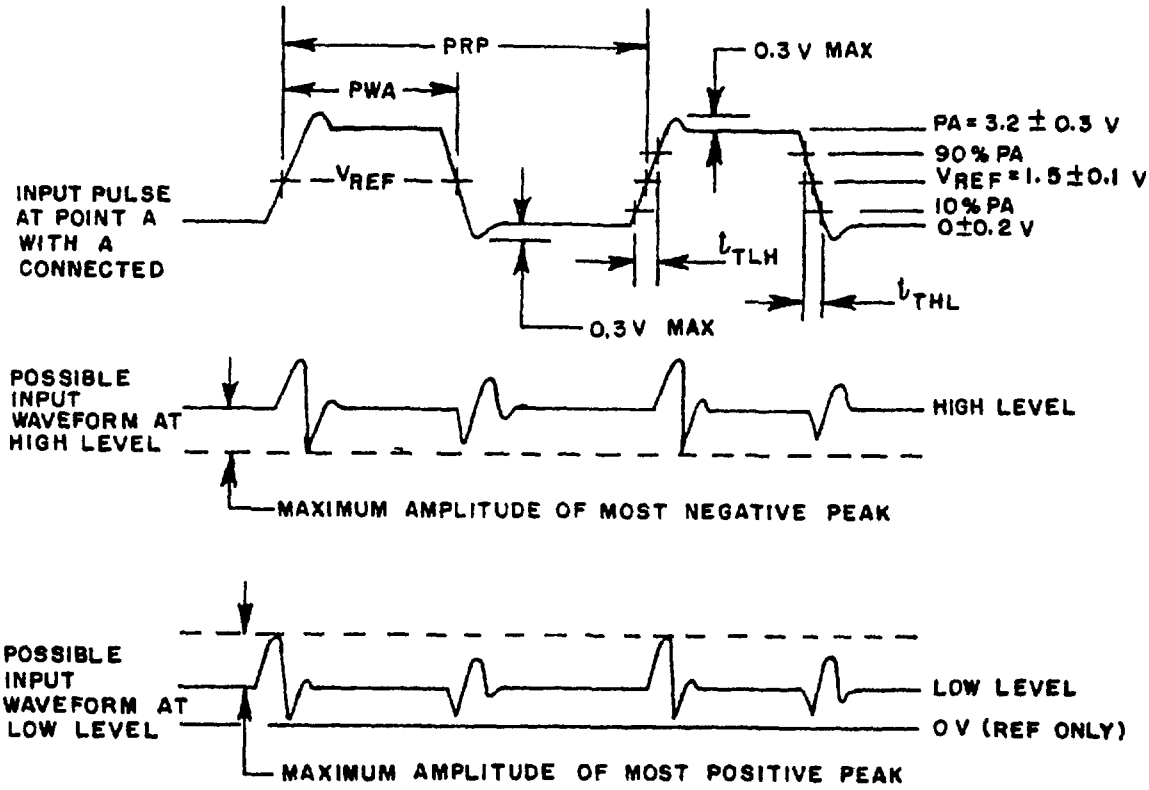
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NOTES:
1. R1, R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±5%, 1/4 W.
3. Load each output with load as specified on figure 18.
4. For test conditions T1-T8, adjust item P01-2 to 0.65 V ±1%.
5. For test conditions T9-T16, adjust item P01-2 to 2.2 V ±1%.
6. C = Sufficient capacitance to reduce item P01-2 power supply noise to < 50 mV.

FIGURE 32. Input crosstalk (low-level and high-level) test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   t_{TLH}, t_{THL} = 20 ±2 ns.
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. The lead length of electrical conductors between terminator resistor R2 and input pins, probes Y01-1, Y01-2 to input pins, load circuits to output pins, and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than two inches (50.8 mm).

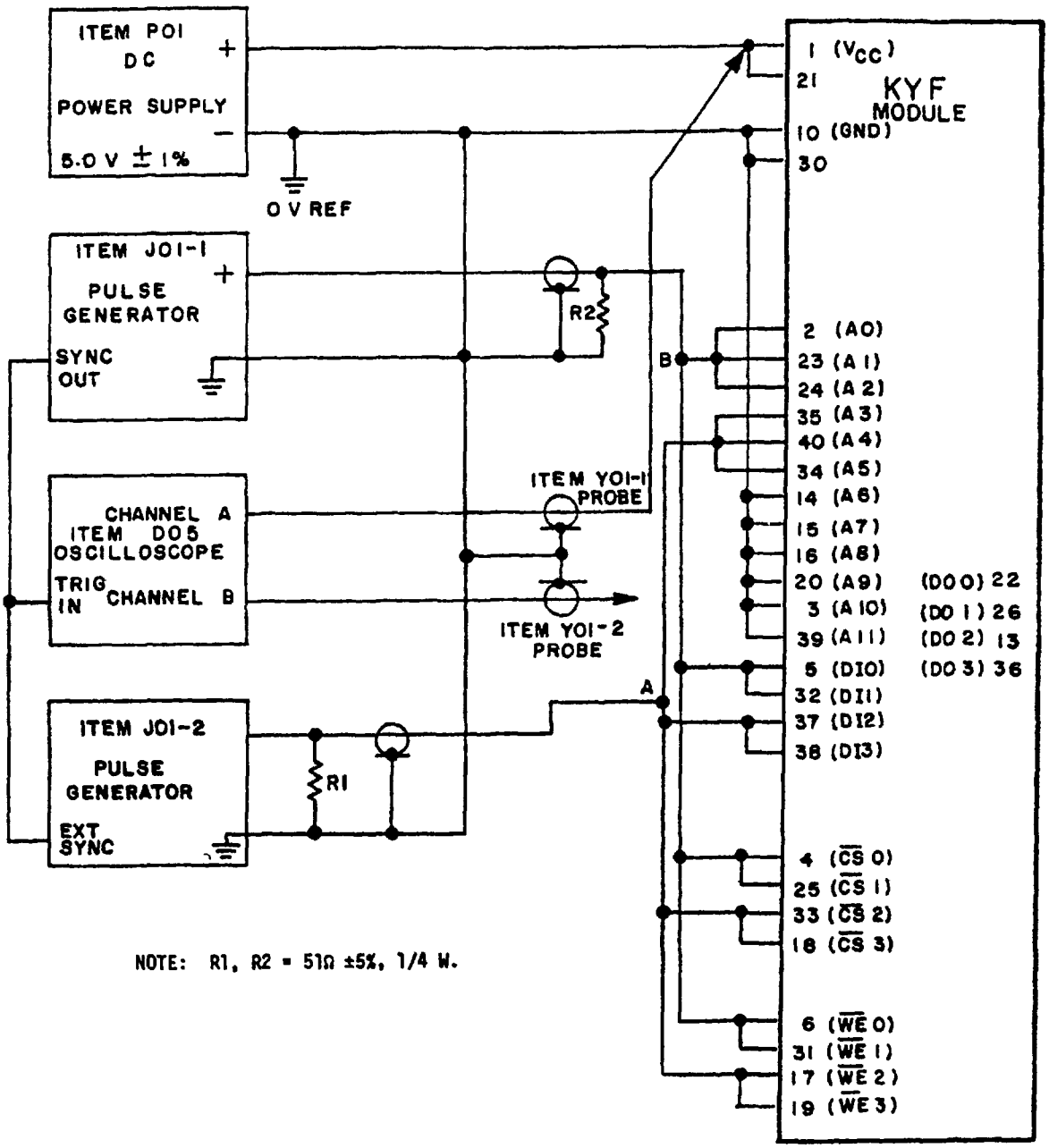
FIGURE 33. Input crosstalk (low-level and high-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	700	mV	34,35

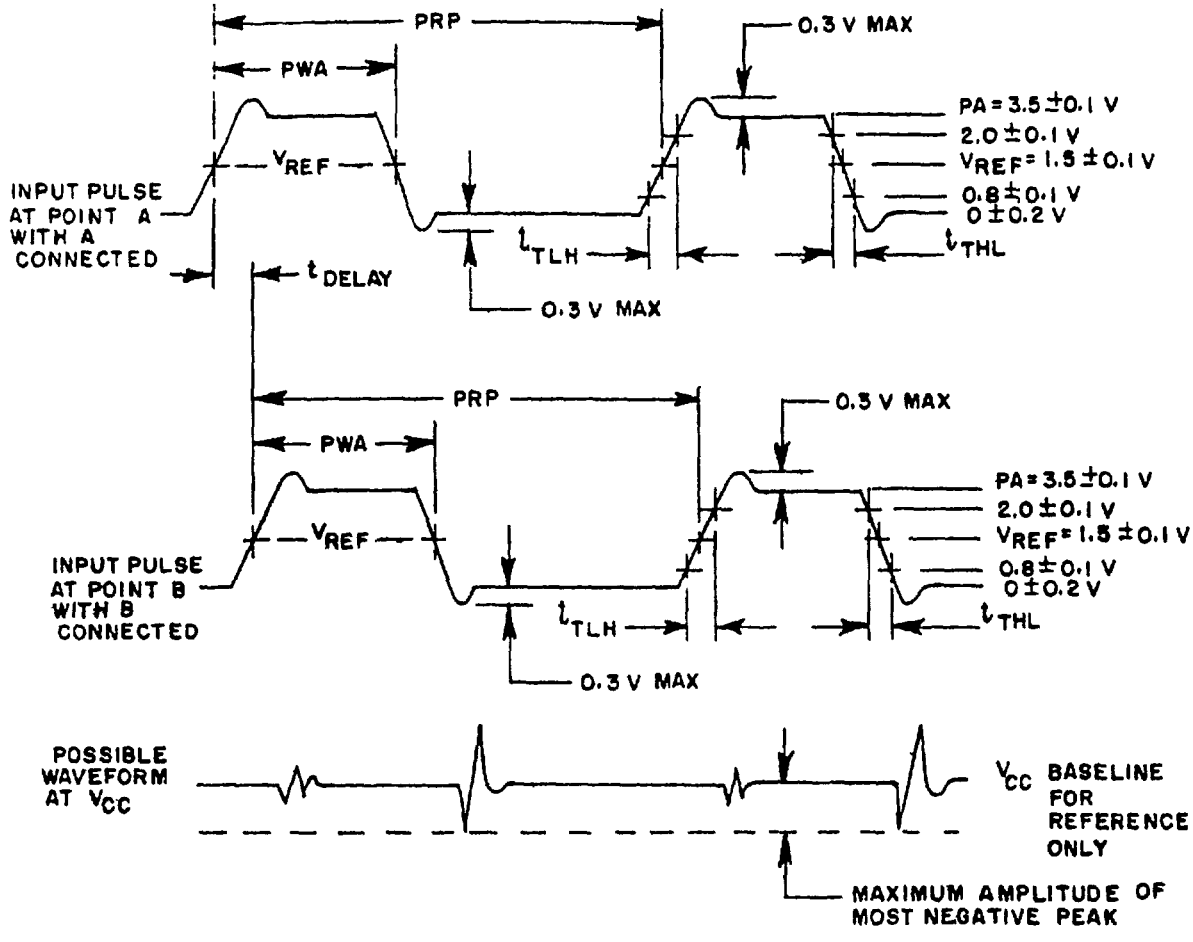
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NOTE: R1, R2 = 51Ω ±5%, 1/4 W.

FIGURE 34. Power supply transient amplitude test circuit for table XIX.

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NOTES:

1. Input pulse parameters for items J01-1 and J01-2:
   t_{TLH}, t_{THL} = 3.5 ±1.5 ns verified at each input pin.
   PWA = 500 ±50 ns.
   PRP = 3,000 ±300 ns.
   t_{DELAY} = 0 ±1 ns.
2. Two items J01 are required due to excessive module loading of the pulse generator.
3. All voltages are referenced to ground.

FIGURE 35. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	3.9	7.3	µF	36

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TABLE XXI. Input and output capacitance test.

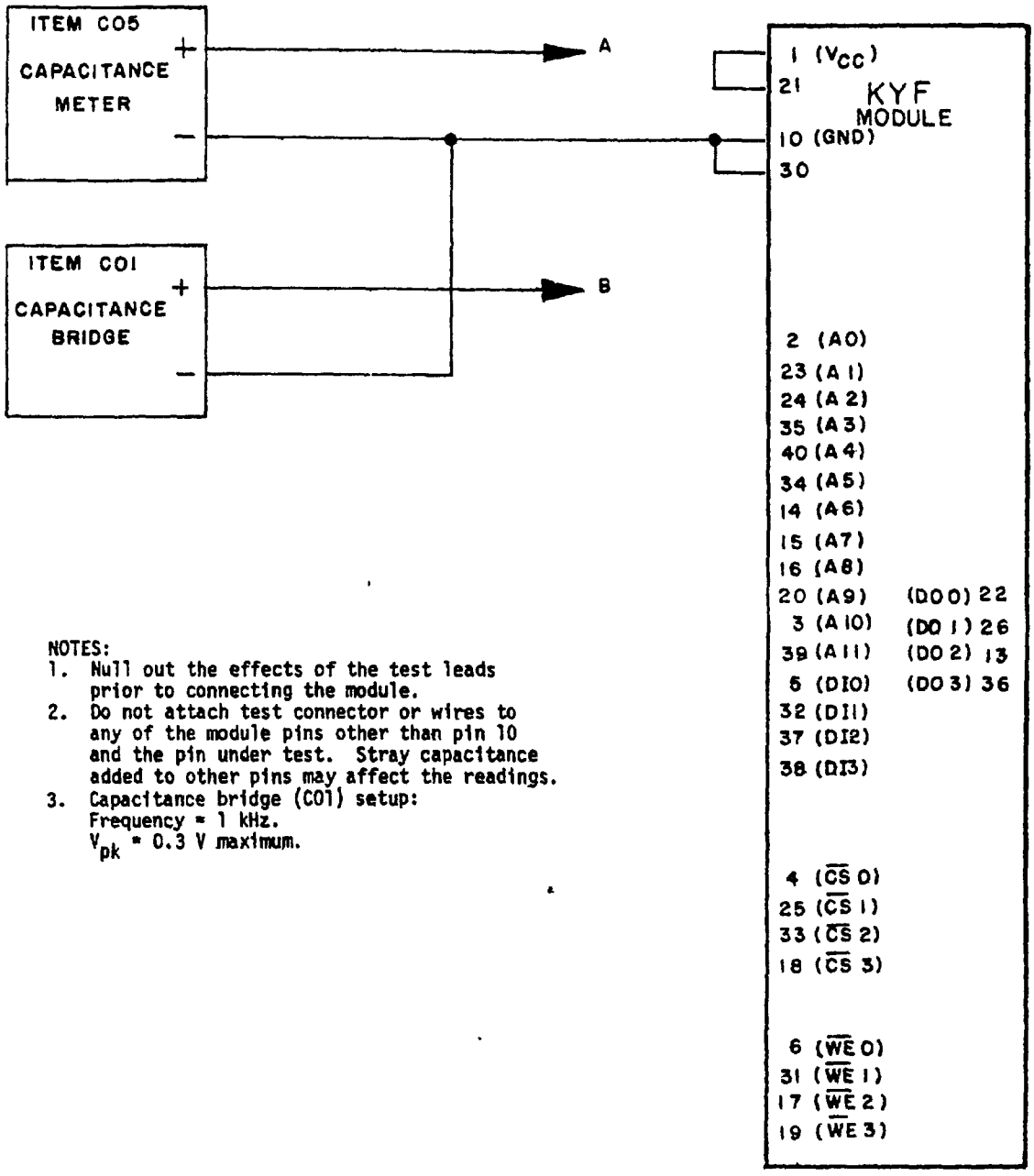
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	$\overline{CS}0$ Input capacitance	4	Open	2.0	25.0	pF	36
W2	$\overline{CS}1$	25					
W3	$\overline{CS}2$	33					
W4	$\overline{CS}3$	18			25.0		
W5	A0	2			45.0		
W6	A1	23					
W7	A2	24					
W8	A3	35					
W9	A4	40					
W10	A5	34					
W11	A6	14					
W12	A7	15					
W13	A8	16					
W14	A9	20					
W15	A10	3					
W16	A11	39			45.0		
W17	$\overline{WE}0$	6			15.0		
W18	$\overline{WE}1$	31			15.0		
W19	$\overline{WE}2$ Input capacitance	17	Open	2.0	15.0	pF	36

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TABLE XXI. Input and output capacitance test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W20	$\overline{WE3}$ Input capacitance	19	Open	2.0	15.0	pF	36
W21	DI0	5					
W22	DI1	32					
W23	DI2	37					
W24	DI3 Input capacitance	38			15.0		
X1	DO0 Output capacitance	22			25.0		
X2	DO1	26					
X3	DO2	13					
X4	DO3 Output capacitance	36	Open	2.0	25.0	pF	36

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 36. Power supply filter capacitance, input and output capacitance test circuit for tables XX and XXI.

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